

LG Energy Solution ESS AZ., Tenant Improvement, Phase I

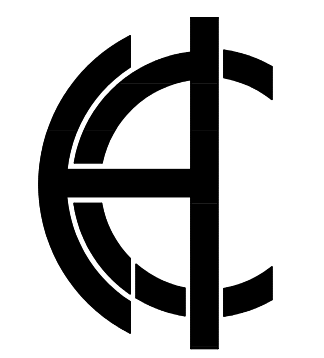
- Truck Court Concrete Repair

West 202 Logistics, 555 and 675 N. 55th Ave., Phoenix, AZ 85043

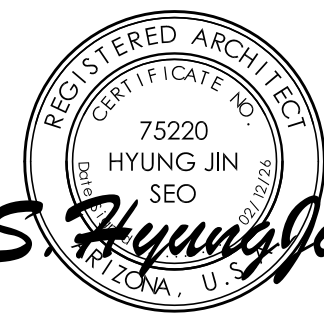
City Submittal - 02/19/2026



LG Energy Solution



HACE INC. 2102 Business Center Dr.
Irvine, CA 92612
1949-BE-2732 L. Howell@hacinc.com



EXPIRES 03/31/28

LG Energy Solution ESS AZ.
Tenant Improvement West 202 Logistics
555 and 675 N. 55th Ave.,
Phoenix, AZ 85043
Bldg. F: 675 N. 55th Ave.
Bldg. G: 555 N. 55th Ave.

Revision	Description	Date

Revision	Description	Date

Key Plan N.T.S.

Date	Drawn By
Project Number	Checked By
202601.05	
Sheet Title	

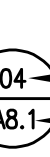
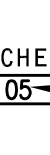
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Sheet Number

A000

ABBREVIATIONS			
&	AND	JAN.	JANITOR
L	ANGLE	JO.	JOINT
Q	AT	KIT.	KITCHEN
Ø	CENTERLINE		
#	DIAMETER OR ROUND	LAB.	LABORATORY
(E)	EXISTING	LAM.	LAMINATE
	FOUND OR NUMBER	LAV.	LAVATORY
		LVR.	LOCKER
		LWT.	LIGHT
ACOUS.	ACOUSTICAL	M.X.	MAXIMUM
A.D.	AREA DRAIN	M.C.	MEDICINE CABINET
AESS.	ARCHITECTURAL	MECH.	MECHANICAL
	ASTHETIC STEEL	MEMB.	MEMBRANE
AGGR.	AGGREGATE	MET.	METAL
ALUM.	ALUMINUM	MFR.	MANUFACTURER
APPROX.	APPROXIMATE	MH.	MANHOLE
ARCH.	ARCHITECT	MIN.	MINIMUM
ASPH.	ASPHALT	MIR.	MIRROR
		MISC.	MISCELLANEOUS
BD.	BOARD	M.O.	MASONRY OPENING
BTUM.	BOTTOM	M.D.	MOUNTED
BIDG.	BUILDING	MTL.	MATERIAL
BLK.	BLOCK	MUL.	MULLION
BKLG.	BLOCKING		
BSM.	BEAM		
BOT.	BOTTOM		
CAB.	CABINET	N.I.C.	NOT IN CONTRACT
C.B.	CATCH BASIN	N.O.	NUMBER
CEM.	CEMENT	NOM.	NOMINAL
CER.	CERAMIC	N.T.S.	NOT TO SCALE
C.G.	CORNER GUARD		
C.I.	CAST IRON	O.A.	OVERALL
C.L.	CULING	OBS.	OBSOLETE
C.M.C.	CHALKING	O.C.	ON CENTER
C.O.	CLOSET	O.D.	OUTSIDE DIAMETER
C.L.P.	CLEAR	O.F.	OFFICE
CNTR.	CENTER	OPNG.	OPENING
C.O.	CASED OPENING	OPP.	OPPOSITE
COL.	COLUMN	O.L.F.	OCCUPANT LOAD FACTOR
CONC.	CONCRETE		
CONN.	CONNECTION	PRCST.	PRE-CAST
CONSTR.	CONSTRUCTION	P. PL.	PL. PLATE
CONT.	CONTINUOUS	P. LAM.	PLASTIC LAMINATE
CORR.	CORRIDOR	PLAS.	PLASTER
CTSK.	COUNTERSINK	PLYWOOD.	PLYWOOD
CTR.	COUNTER	PAIR	PAIR
	DOUBLE	P.T.	POINT
DBL.	DOUBLE	P.T.D.	PAPER TOWEL DISPENSER COMBINATION
D.C.	DEPARTMENT	P.T.D.	PAPER TOWEL DISPENSER & RECEPTION
DET.	DETAIL	P.T.R.	PARITION
D.F.	DRINKING FOUNTAIN		
D.I.	DIAMETER		
DM.	DIMENSION		
D.M.	DIMENSION		
DSP.	DOW	Q.	QUARRY TILE
D.	DOWN		
D.O.	DOOR OPENING	R.	RISER
DR.	DOOR	R.D.	RADIUS
D.S.	DOWNPOUT	R.O.	ROOF DRAIN
D.S.P.	DRY STANDPIPE	REF.	REFERENCE
DWG.	DRAWING	REFR.	REFRIGERATOR
DWR.	DRAWER	REG.	REGISTER
		REIN.	REINFORCED
		REQ.	REQUIRED
E.A.	EACH	RESIL.	RESILIENT
E.D.F.	ELECTRICAL DRINKING FOUNTAIN	R.F.	ROUGH
E.J.	EXPANSION JOINT	R.O.	ROUGH OPENING
EL.	ELEVATION	RWD.	RAIN WATER LEADER
ELEC.	ELECTRIC	R.W.L.	RAIN WATER LEADER
ELEV.	ELEVATOR		
EMER.	EMERGENCY	S.C.	SOLID CORE
ENCL.	ENCLOSURE	S.C.D.	SEAT COVER DISPENSER
E.P.	ELECTRICAL PANELBOARD	SCH.	SCHEDULE
EQ.	EQUAL	S.D.	SOAP DISPENSER
EQPT.	EQUIPMENT	SH.	SHELL
E.W.C.	ELECTRICAL WATER COOLER	SECT.	SECTION
EXST.	EXISTING	SH.	SHOWER
EXP.	EXPANSION	SHT.	SHEET
EXPD.	EXPOSURE	SHR.	SIMILAR
EXT.	EXTERNAL	S.N.C.	SANITARY NAPKIN DISPENSER
		S.N.D.	SANITARY NAPKIN RECEPTACLE
F.A.	FIRE ALARM	SPEC.	SPECIFICATION
F.B.	FLAT BAR	S.Q.	SQUARE
F.D.	FLOOR DRAIN	S.S.	STAINLESS STEEL
FDN.	FOUNDATION	S.S.K.	SERVICE SINK
F.E.	FIRE EXTINGUISHER	STA.	STATION
F.E.C.	FIRE EXTINGUISHER CABINET	STD.	STANDARD
F.F.C.	FIRE FIGHTER	STL.	STEEL
F.H.C.	FIRE HOSE CABINET	STR.	STORAGE
FIN.	FINISH	STR.	STRUCTURAL
FL.	FLOOR	SUSP.	SUSPENDED
FLASH.	FLASHING	SYM.	SYMMETRICAL
FLUOR.	FLUORESCENT		
F.O.C.	FACE OF CONCRETE	T.B.	TREAD
F.O.F.	FACE OF FINISH	T.C.	TOWEL BAR
F.O.M.	FACE OF MASONRY	T.O.	TOP OF CURB
F.O.S.	FACE OF SLAB	T.P.	TOILET PARAPET
FRF.	REFRIGERATOR	T.R.	TERRAZZO
F.R.	FIRE-RATED	T&G.	TONGUE AND GROOVE
F.S.	FULL SIZE	TRK.	TRUCK
FT.	FOOT OR FEET	T.O.C.	TOP OF CANOPY
FTG.	FOOTING	T.O.S.	TOP OF MECHANICAL SCREEN
FURR.	FURRING	T.O.P.	TOP OF PARAPET
FUTE.	FUTURE	T.O.S.	TOP OF SURFACE
		T.O.W.	TOP OF WALL
GA.	GALVANIZED	T.P.	TOP OF PAVEMENT
GALV.	GALVANIZED	T.P.	TOILET PARAPET DISPENSER
G.B.	GRAB BAR	T.V.	TELEVISION
G.I.S.M.	GALVANIZED IRON SHEET METAL	TYP.	TYPICAL
GL.	GLASS		
GRD.	GROUND		
GR.	GRADE	U.O.N.	UNLESS OTHERWISE NOTED
GTP.	GRATE		
		VERT.	VERTICAL
H.B.	HOLE BIB	VEST.	VESTIBULE
H.C.	HOLLOW CORE		
H.WD.	HARDWOOD	W/	WITH
HOME.	HARDWARE	W.C.	WATER CLOSET
H.M.	HOLLOW METAL	WD.	WOOD
HORIZ.	HORIZONTAL	W/O.	WITHOUT
H.R.	HOUR	W.P.	WATERPROOF
HGT.	HEIGHT	WT.	WEIGHT
LD.	INSIDE DIAMETER		
IN.	INCH OR INCHES		
INSUL.	INSULATION		
INT.	INTERIOR		

SYMBOLS









COLUMN LINE

SECTION

SECTION IDENTIFICATION

SHEET WHERE SECTION IS DRAWN

ELEVATION

ELEVATION IDENTIFICATION

SHEET WHERE ELEVATION IS DRAWN

DETAIL

DETAIL IDENTIFICATION

SHEET WHERE DETAIL IS DRAWN

INTERIOR ELEVATIONS

ELEVATION IDENTIFICATION

SHEET WHERE ELEVATION IS DRAWN

ROOM NUMBER

ROOM NAME

ROOM NUMBER

REVISION

CLOUD AROUND REVISED AREA

REVISION NUMBER

DATUM POINT

KEYNOTE

ALIGN

SHADED AREA

DENOTES MILLWORK

GENERAL CONSTRUCTION NOTES

1. THE CONTRACTOR AND MATERIALS SHALL BE AS SPECIFIED AND IN ACCORDANCE WITH ALL APPLICATION CODES, ORDINANCES, LAWS, PERMITS, AND THE CONTRACT DOCUMENTS.
2. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE ACCURATE PLACEMENT OF ALL NEW CONSTRUCTION ON THE SITE.
3. THE CONTRACTOR IS RESPONSIBLE FOR ALL EXISTING SURVEY MONUMENTS WHICH MAY BE DISTURBED THROUGH CONSTRUCTION ACTIVITIES DEMAND RECOVERY. IF ANY MONUMENTS ARE LOST OR DESTROYED, THE APPLICANT SHALL HAVE A LICENSED LAND SURVEYOR QUALIFIED REGISTERED CIVIL ENGINEER REESTABLISH ANY SUCH MONUMENTATION DAMAGED OR DESTROYED DURING THE CONSTRUCTION OF THE PROJECT AND SUBMIT "PRELIMINARY CORNER RECORDS" TO THE COUNTY ENGINEER AFTER REPLACEMENT. AFTER APPROVAL BY THE CIVIL ENGINEER, THE APPLICANT SHALL FILE THE CORNER RECORDS WITH THE COUNTY SURVEYOR. PRIOR TO THE EXEMPTION OF ANY SECURITY, EVIDENCE OF SUCH FILING SHALL BE FURNISHED TO THE CIVIL ENGINEER.
4. THE CONTRACTOR IS RESPONSIBLE TO ENSURE THAT TRUCK ROUTES SHALL BE STEERED AWAY FROM RESIDENTIAL AREAS. (SEE CHAPTER 4)
5. THE CONTRACTOR SHALL ENSURE THAT TRUCKS HAULING DIRT ON PUBLIC ROADS TO PREPARE THE SITE SHALL COVERED CHASSIS MAINTAIN A 4" DIFFERENTIAL BETWEEN THE MAXIMUM HEIGHT OF ANY HAULED MATERIAL AND THE TOP OF THE HAUL TRAILER. HAUL TRUCK DRIVERS SHALL WATER THE LOAD PRIOR TO LEAVING THE SITE TO PREVENT SOIL LOSS DURING TRANSPORT.
6. THE CONTRACTOR SHALL ENSURE THAT GRADED SURFACES USED FOR OFF ROAD PARKING, FILL PAVED AND GRADED TRAFFIC CONTROL DEVICES SHALL BE STABILIZED FOR DUST CONTROL THAT NEEDED. FREQUENTLY ACCESSED UNPAVED AREAS SHALL BE PAVED AS EARLY AS POSSIBLE TO MINIMIZE DIRT TRACKOUT BY PUBLIC RIGHTS-OF-WAY.
7. THE CONTRACTOR SHALL COORDINATE ANY LINE INTERFERENCES OR DETOURS WITH THE CITY, FILL PAVED AND GRADED TRAFFIC CONTROL DEVICES SHALL BE USED AS NEEDED TO MAINTAIN CONSTRUCTION ACTIVITY INTERFERENCES WITH OFF-SITE TRAFFIC.
8. THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND SITE CONDITIONS BEFORE STARTING WORK. SHOULD A DISCREPANCY APPEAR IN THE CONTRACT DOCUMENTS, OR BETWEEN THE CONTRACT DOCUMENTS AND EXISTING CONDITIONS, NOTIFY THE ARCHITECT AT ONCE FOR INSTRUCTION ON HOW TO PROCEED.
9. SHOULD A CONFLICT OCCUR BETWEEN DRAWINGS AND SPECIFICATIONS, THE MORE RESTRICTIVE CONDITION SHALL TAKE PRECEDENCE, UNLESS A WRITTEN DECISION OF THE ARCHITECT HAS BEEN OBTAINED WHICH DESCRIBES A CLARIFICATION OR ALTERNATE METHOD AND/OR MATERIALS.
10. THE CONTRACTOR SHALL CONFINING HIS OPERATIONS ON THE SITE TO AREAS PERMITTED BY OWNER.
11. THE JOB SITE SHALL BE MAINTAINED IN A CLEAN, ORDERLY CONDITION FREE OF DEBRIS AND LITTER. EACH SUBTRACTOR SHALL NOT BE UNREASONABLY ENCUMBERED WITH ANY MATERIALS OR EQUIPMENT. EACH SUBTRACTOR IMMEDIATELY UPON COMPLETION OF EACH PHASE OF THE WORK SHALL REMOVE ALL TRASH AND DEBRIS AS A RESULT OF HIS OPERATION.
12. ALL MATERIAL STORED ON THE SITE SHALL BE PROPERLY STACKED AND PROTECTED TO PREVENT WEAR AND DEGRADATION. FAILURE TO PROTECT MATERIALS MAY CAUSE FOR REJECTION OF WORK.
13. THE CONTRACTOR SHALL DO ALL CUTTING, FITTING, OR PATCHING OF HIS WORK THAT MAY BE REQUIRED TO MAKE ITS SEVERAL PARTS FIT TOGETHER PROPERLY AND SHALL NOT EXCHANGE ANY OTHER WORK BY CUTTING, EXCHANGING, OR OTHERWISE ALTERING THE WORK. THE CONTRACTOR SHALL NOT REMOVE OR NOT ANY OBSERVED DEFICIENCIES HAVE TO BE CORRECTED TO CONFORM TO THE APPROVED PLANS AND SPECIFICATIONS.
14. STRUCTURAL SURROUNDING BY THE ENGINEER/ARCHITECT SHALL BE PERFORMED. A STATEMENT IN WRITING SHALL BE GIVEN TO THE BUILDING OFFICIAL, STATING THAT THE WORK IS BEING DONE BY THE CONTRACTOR IN ACCORDANCE WITH THE APPROVED PLANS AND SPECIFICATIONS TO CONFORM TO THE APPROVED PLANS AND SPECIFICATIONS.
15. PROVIDE FIRE EXTINGUISHERS PER THE REQUIREMENTS OF LOCAL GOVERNING AGENCIES.
16. PROVIDE ALL ACCESS PLACES AS REQUIRED BY GOVERNING CODES TO ALL CONCEALED SPACES, VOIDS, ATTICS, ETC. VERIFY TYPE REQUIRED WITH ARCHITECT PRIOR TO INSTALLATION.
17. NO PORTION OF WORK REQUIRING A SHOP DRAWING OR SAMPLE SUBMISSION SHALL BE COMMENCED UNTIL THE SUBMISSION HAS BEEN REVIEWED BY THE ARCHITECT. ALL SUCH PORTIONS OF THE WORK SHALL BE IN ACCORDANCE WITH CORRECTED SHOP DRAWINGS AND SAMPLES.
18. DIMENSIONS:
 - A. ALL DIMENSIONS SHALL TAKE PRECEDENCE OVER SCALE.
 - B. ALL DIMENSIONS ARE TO THE FINISH UNLESS OTHERWISE NOTED.
 - C. CEILING HEIGHT DIMENSIONS ARE FROM FINISHED FLOOR TO TOP OF FACE OF FINISH CEILING MATERIAL UNLESS OTHERWISE NOTED.
19. DO NOT SCALE DRAWINGS.
20. PROVIDE ALL NECESSARY BLOCKING, BACKING, AND FRAMING FOR LIGHT FIXTURES, ELECTRIC UNITS, A.C. EQUIPMENT, RECESSED ITEMS, AND ALL OTHER ITEMS AS REQUIRED.
21. WHERE LARGER STUDIOS OR FURNING ARE REQUIRED TO COVER PIPING AND CONDUITS, THE LARGER STUDIO SIZE OR FURNING SHALL ENCLOSE THE FULL SURFACE OF THE WALL WITHIN AND LENGTH WHERE THE FURNING OCCURS.
22. ALL LEGAL EXIT DOORS SHALL BE OPERABLE FROM THE INSIDE WITHOUT THE USE OF A KEY OR SPECIAL KNOWLEDGE OR EFFORT. EXIT SIGNS SHALL BE PROVIDED AT ALL EXITS EXCEPT BY THE FIRE DEPARTMENT SECTION 101.1. ALL DOOR SWINGS SERVING AN OCCUPANT LOAD OF 50 OR GREATER SHALL SWING IN THE DIRECTION OF TRAVEL.
23. ALL GLASS AND GLAZING SHALL COMPLY WITH IBC CHAPTER 24.
24. THE LINE LOADS FOR WHICH EACH FLOOR OR PORTION THEREOF OF A COMMERCIAL OR INDUSTRIAL BUILDING SHALL BE DESIGNED SHALL HAVE SUCH DESIGN LINE LOADS COMPLYING WITH THE REQUIREMENTS OF THE IBC SECTION 1607.5.5 WHICH THEY APPLY, USING DURABLE METAL SIGNS, AND IT SHALL BE UNLAWFUL TO REMOVE OR DEFACE SUCH NOTICES. THE OCCUPANT OF THE BUILDING SHALL BE RESPONSIBLE FOR KEEPING THE LINE LOADS BELOW THE ALLOWABLE LIMITS. IBC SECTION 1607.5.5.
25. IN CASE OF ANY DISCREPANCIES CONTACT THE A.O.R. THE MORE STRINGENT CRITERIA SHALL ALWAYS GOVERN.
26. A PREDEMOITION MEETING WITH THE INSPECTION DEPARTMENT IS REQUIRED PRIOR TO START OF DEMOLITION. CALL (802) 262-5510 TO SCHEDULE AN INSPECTION.
27. SPRINKLER AND/OR ALARM SYSTEM PLANS MUST BE REVIEWED AND APPROVED BY THE PHASED BUILDING AUTHORITY PRIOR TO ANY SPRINKLER ALARM SYSTEM MODIFICATION OR REPLACEMENT.

APPLICABLE CODES AND STANDARDS

INTERNATIONAL BUILDING CODE, 2024 EDITION WITH PHOENIX AMENDMENT
INTERNATIONAL MECHANICAL CODE, 2024 EDITION WITH PHOENIX AMENDMENT
NATIONAL ELECTRICAL CODE, 2024 EDITION WITH PHOENIX AMENDMENT
UNIFORM PLUMBING CODE, 2024 EDITION WITH PHOENIX AMENDMENT
INTERNATIONAL ENERGY CONSERVATION CODE, 2024 EDITION
INTERNATIONAL FIRE CODE, 2018 EDITION WITH PHOENIX AMENDMENT

PLANNING NOTES

- LEGAL JURISDICTION: CITY OF PHOENIX
- ZONING CATEGORY: A-1 AND A-2 (LIGHT AND HEAVY INDUSTRIAL)
- VILLAGE: ESTRELLA

CODE ANALYSIS

CONSTRUCTION TYPE:	III-B 400'-0" TRAVEL DISTANCE
AUTOMATIC FIRE SPRINKLERS:	YES
OCCUPANCY TYPE:	B, S-1, AND F-1 GROUP NON SEPARATED MIXED USE
GROSS BUILDING AREA:	BLDG. F: 301,711 SF, UNLIMITED AREA BLDG. G: 295,586 SF, UNLIMITED AREA
NUMBER OF STORIES:	1
FIRE RESISTANCE RATED ASSEMBLIES CONSTRUCTED OR ALTERED IN TYPE OF WORK:	
STRUCTURAL FRAME:	0
EXTERIOR WALLS (BEARING):	2
INTERIOR WALLS (BEARING):	0
EXTERIOR WALLS (NONBEARING):	0
INTERIOR WALLS (NONBEARING):	0
FLOOR CONSTRUCTION:	0
ROOF CONSTRUCTION:	0
PARKING:	NO CHANGE BLDG. F: 208 SPACES (7 ACCESSIBLE STALLS) BLDG. G: 197 SPACES (6 ACCESSIBLE STALLS)

SHEET INDEX

	DATE ISSUED	DWGS IN THIS SET	
			
			
			
			
			ARCHITECTURAL
			CVR COVER SHEET
			A001 GENERAL INFORMATION & SHEET INDEX
			A002 SPECIFICATIONS
			A101 SITE PLAN
			STRUCTURAL
			S901 STRUCTURAL GENERAL NOTES
			S501 TYPICAL CONCRETE DETAILS
			S1.10 CONCRETE SLAB PLAN

PROJECT DIRECTORY

OWNER
CRP LDF 55TH AVENUE LLC (C/O CAPROCK PARTNERS)
1300 DOVE STREET SUITE 200 NEWPORT BEACH, CA 92660
CONTACT: RICK GOEBEL PH: 702-501-4906
E-MAIL: RGOEBEL@CAPROCK-PARTNERS.COM

TENANT
LG ENERGY SOLUTION,
675 N. 55TH AVE., PHOENIX, AZ 85043
CONTACT: JUNG HO LEE PH: 011-82-10-7187-6127

GENERAL CONTRACTOR
SUN STATE BUILDERS,
1050 W WASHINGTON ST #214, TEMPE, AZ 85288
CONTACT: MIKE FORST PH: 480-894-1286
E-MAIL: MFORST@SUNSTATEBUILDERS.COM

ARCHITECT
HACE INC
2102 BUSINESS CENTER DR., IRVINE, CA 92612
CONTACT: HYUNG JIN SEO PH: 949-892-9732
E-MAIL: HSEO@HACEINC.COM

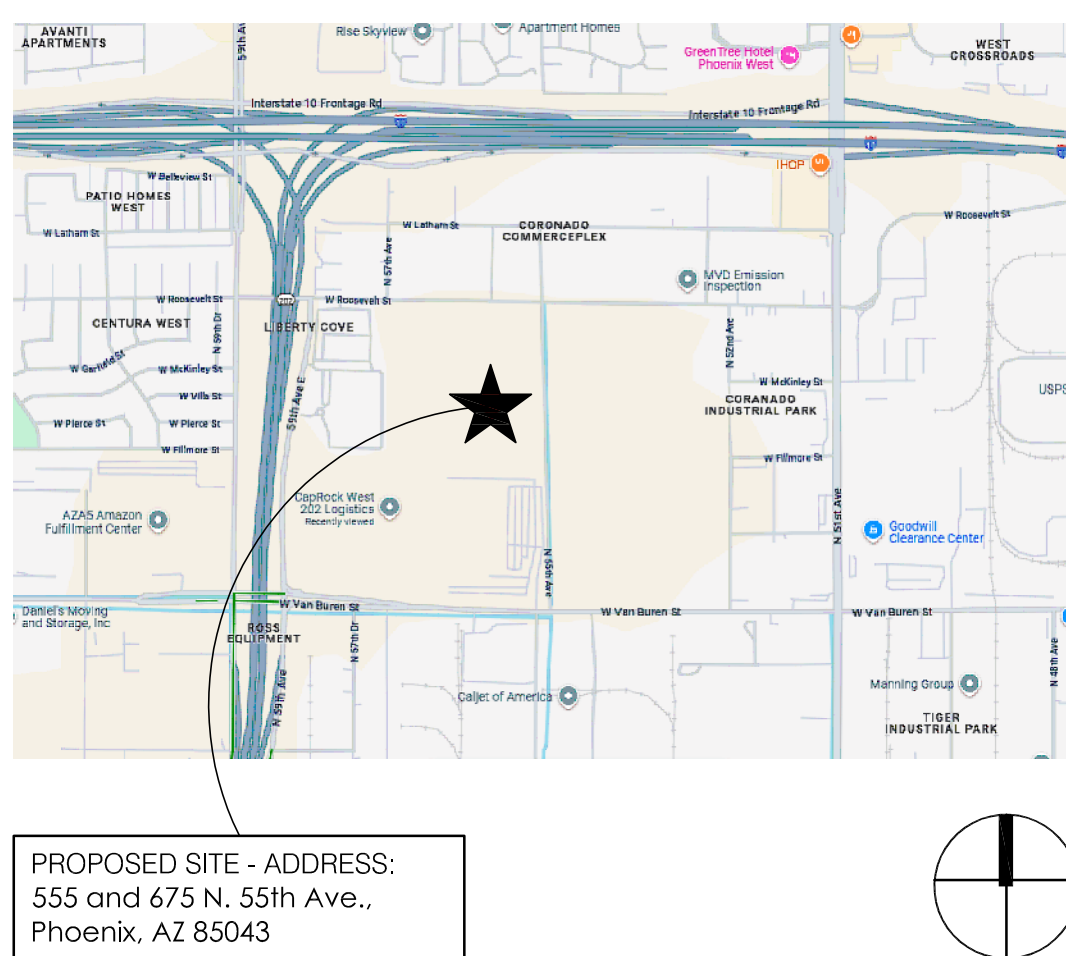
STRUCTURAL
BURKE STRUCTURAL ENGINEERS
151 KALMUS DR, STE E140, COSTA MESA, CA 92626
CONTACT: MOHAMED BADAWI PH: 657-289-0460
E-MAIL: MOHAMED@BURKESE.COM

ELECTRICAL, MECHANICAL, AND PLUMBING ENGINEER
3S MEP+S
 27475 FERRY ROAD WARRENVILLE, IL 60555
 CONTACT: SUSHIL KUMAR P.E. PH: 312-446-3327
 E-MAIL: SUSHIL@3SMEP.COM

SCOPE OF WORK

INTERIOR TENANT IMPROVEMENT FOR LGES PHASE I. THE SCOPE OF WORK INCLUDES THE REMOVAL OF THE EXISTING TRUCK COURT CONCRETE AND THE INSTALLATION OF NEW HEAVY-DUTY CONCRETE, ALONG WITH ASSOCIATED ELECTRICAL AND PLUMBING LINES TO SUPPORT FUTURE WATER TESTING EQUIPMENT AND DATA CENTER OPERATIONS.

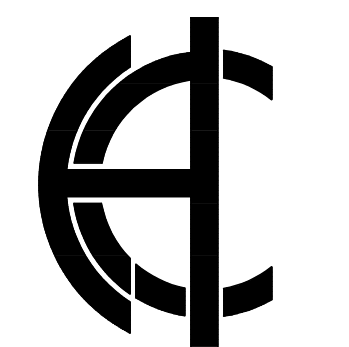
VICINITY MAP



DEFERRED SUBMITTALS

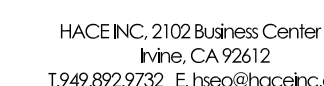
THIS PROJECT HAS BEEN PERMITTED WITHOUT REVIEW AND/OR APPROVAL OF THE FOLLOWING DEFERRED SUBMITTALS. PLANS APPROVED BY THE TOWNSHIP SHALL BE OBTAINED FOR EACH DEFERRED ITEM LISTED BELOW PRIOR TO COMMENCING ANY WORK WITHIN THE SCOPE OF SUCH DEFERRAL. DEFERRALS MUST BE REVIEWED AND ACCEPTED BY THE ARCHITECT OR ENGINEER OF RECORD PRIOR TO SUBMITTING FOR REVIEW WITH THE CITY.

FIRE PROTECTION
FIRE ALARM

[illegible]

Revised	Description	Date
-	CITY SUBMITTAL/ISSUE #40.35F	09/12/26

Date	Drawn By
Project Number 202601.05	Checked By
Sheet Title	



Revisão	Description	Data
001/2020	CPI-LSM-FUJ/2018-NC-LSI	

Equi Flow	NTS
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Date	Drawn By
Project Number 202601.05	Checked By
Sheet Title	

SITE PLAN

Sheet Number

A101



REF. SCALE: 1" = 40'-0"

LEGEND

- 5 REMOVE EXISTING CONCRETE RAMP, AND PREP. FOR NEW PLATFORM (SEPARATE PERMIT)
- 6 750 KW GENERATOR (SEPARATE PERMIT)
- 7 INSTALL 1" CONDUIT UNDERGROUND FOR POWER

1. INSTALL MIN. 4" CONDUIT UNDERGROUND FOR FUTURE DATA CENTER AND IDF
2. INSTALL 2" CONDUIT UNDERGROUND FOR DATA
3. INSTALL 1/2" CONDUIT UNDERGROUND FOR FUTURE WATER TEST
4. INSTALL 3/4" DIA. DOMESTIC WATER PIPE UNDERGROUND

KEYNOTES

GENERAL STRUCTURAL NOTES:

- COORDINATION: THE CONTRACTOR SHALL VERIFY ALL DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL BE RESPONSIBLE FOR COORDINATION OF ALL WORK AND MATERIALS INCLUDING THOSE FURNISHED BY SUBCONTRACTORS.
2. DISCREPANCIES: THE CONTRACTOR SHALL INFORM THE ENGINEER IN WRITING, OF ANY DISCREPANCIES OR OMISSIONS NOTED ON THE DRAWINGS THAT DO NOT SUBMIT TO CORRECTION. IN THE EVENT OF DISCREPANCIES UPON RECEIPT OF SUCH INFORMATION, THE ENGINEER WILL SEND INSTRUCTIONS TO ALL CONCERNED. ANY SUCH DISCREPANCY, OMISSION, OR VARIATION NOT REPORTED SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR.
3. TYPICAL DETAILS AND GENERAL STRUCTURAL NOTES SHALL APPLY, UNLESS SPECIFICALLY SHOWN OTHERWISE. ANY CONFLICTS OR DETAIL NOT FULLY SHOWN OR NOTED SHALL BE SIMILAR TO DETAILS SHOWN FOR SIMILAR CONDITIONS. ALL CONSTRUCTION WORK SHALL COMPLY WITH ALL APPLICABLE BUILDING CODES, REGULATIONS AND SAFETY REQUIREMENTS.
4. TRADE NAMES: WHERE AN ITEM IS IDENTIFIED BY A TRADE NAME THE SUFFIX "OR APPROVED EQUIVALENT" SHALL BE IMPLIED UNLESS SPECIFICALLY NOTED OTHERWISE.
5. STANDARDS: EXCEPT WHERE MORE STRINGENT REQUIREMENTS ARE NOTED OR SPECIFIED, THE PLANS OR SPECIFICATIONS, ALL PHASES OF WORK SHALL CONFORM TO THE MINIMUM STANDARDS OF THE **2024 INTERNATIONAL BUILDING CODE (IBC)**.
6. INSPECTION: WHEREVER SPECIAL AND/OR CONTINUOUS INSPECTION IS CALLED FOR IN THESE NOTES OR THE BUILDING CODES, SUCH INSPECTION SHALL BE PERFORMED BY A TESTING LABORATORY EMPLOYED BY THE OWNER. JOB SITE VISITS BY THE ENGINEER SHALL NOT CONSTITUTE AN OFFICIAL INSPECTION.
7. PRE-CONSTRUCTION MEETING: PRIOR TO THE COMMENCEMENT OF THE CONSTRUCTION, THE ARCHITECT SHALL CONDUCT A MEETING TO ESTABLISH COMMUNAL ROUTES, ACCESS AND SEATING REQUIREMENTS FOR THE CONSTRUCTION ADMINISTRATION OF THE PROJECT, REPRESENTATIVES FROM THE ENGINEER, TESTING LABORATORY, BUILDING DEPARTMENT, CONTRACTOR, AND DESIGN PROFESSIONALS AS WELL AS APPROPRIATE WILL ATTEND. ("IF NECESSARY")
8. OTHER TRADES: SEE ARCHITECTURAL AND CONSULTANT DRAWINGS FOR SIZE AND LOCATION OF PIPE, OTHER OPENINGS, ANCHOR BOLT REQUIREMENTS FOR EQUIPMENT AND OTHER DETAILS NOT SHOWN ON THESE STRUCTURAL DRAWINGS. ALL DIMENSIONS ARE TO BE CHECKED AND VERIFIED WITH THE ARCHITECTURAL DRAWINGS.
9. MATERIALS AND WORKSMANSHIP: THE CONTRACTOR SHALL SUPPLY ALL LABOR, MATERIALS, EQUIPMENT, TOOLS, AND UTILITIES IN ACCORDANCE WITH THE CODES AND POWER, NECESSARY FOR THE PROPER EXECUTION OF THE WORK SHOWN OR INDICATED ON THESE DRAWINGS. ALL MATERIAL SHALL BE NEW MATERIALS. SUBCONTRACTORS SHALL BE SKILLED IN THEIR TRADE.
10. MATERIALS AND WORKSMANSHIP WARRANTY: THE CONTRACTOR SHALL REPLACE ANY DEFECTIVE MATERIALS AND CORRECT POOR WORKSMANSHIP WITH AN ADDITIONAL COST TO THE OWNER. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE MATERIALS OR WORKSMANSHIP WHICH APPEAR IN ONE YEAR FROM THE DATE OF COMPLETION OF THE PROJECT. THIS WARRANTY APPLIES TO THE WORK DONE BY THE SUBCONTRACTORS AS WELL AS THE WORK DONE BY THE CONTRACTOR.
11. SAFETY: THE CONTRACTOR SHALL ADEQUATELY PROTECT HIS WORK, ADJACENT PROPERTY, THE PUBLIC, AND BE RESPONSIBLE FOR DAMAGE OR INJURY DUE TO HIS ACT OR NEGLECT.
12. SIDEWALK PROTECTION: PEDESTRIAN TRAFFIC SHALL BE PROTECTED AS SPECIFIED IN SECTION 3306 OF THE BUILDING CODE.
13. SHORING: IT SHALL BE THE CONTRACTOR'S SOLE RESPONSIBILITY TO DESIGN AND PROVIDE ADEQUATE SHORING, BRACING AND FORMWORK, ETC., AS REQUIRED FOR THE PROTECTION OF LIFE AND PROPERTY DURING THE CONSTRUCTION OF THIS BUILDING.
14. EXCAVATION: THE CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR ALL EXCAVATION PROCEDURES INCLUDING LAGGING, SHORING, AND PROTECTION OF ADJACENT PROPERTY, STRUCTURES, STREETS AND UTILITIES IN ACCORDANCE WITH THE STANDARDS OF THE CITY OF NEWPORT BEACH AND WITH THE JOINING PROPERTY, THE PREMISES SHALL BE KEPT FROM ACCUMULATION OF WASTE MATERIALS, AND THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE COLLECTION AND REMOVAL OF RUBBISH, SURPLUS OF MATERIALS, AND TOOLS AND LEAVE THE BUILDING BROOM CLEAN.
15. SHOP DRAWINGS: SHOP DRAWINGS ARE AN AID FOR FIELD PLACEMENT AND ARE SUPERSEDED BY THE STRUCTURAL DRAWINGS. IT SHALL BE THE RESPONSIBILITY OF THE GENERAL CONTRACTOR TO MAKE CERTAIN CONSTRUCTION IS IN FULL AGREEMENT WITH THE LATEST CONSTRUCTION DOCUMENTS.
16. SHOP DRAWING CHECK: THE REVIEW OF SHOP DRAWINGS BY THE STRUCTURAL ENGINEER IS ONLY FOR GENERAL COMPLIANCE WITH THE STRUCTURAL DRAWINGS AND THE CONSTRUCTION DOCUMENTS. THE CONTRACTOR DOES NOT GUARANTEE THAT THE SHOP DRAWINGS ARE CORRECT NOR DOES IT IMPLY THAT THEY SUPERSEDE THE STRUCTURAL DRAWINGS.
17. SUBSTITUTIONS: BEFORE SUBSTITUTIONS FOR ANY MATERIALS OR SYSTEMS SHOWN ON THE DRAWINGS OR CALLED OUT IN THE SPECIFICATIONS WILL BE CONSIDERED, THE PERSON PROPOSING THE SUBSTITUTION SHALL SUBMIT A WRITTEN LETTER TO THE ARCHITECT AND THE ENGINEER. THE LETTER SHALL BE SIGNED BY THE ARCHITECT, A THE PROPOSER AGREES TO PAY THE ENGINEER FOR THE COST OF EVALUATING THE PROPOSED CHANGE.
- B. THE CONTRACTOR SHALL PAY THE ENGINEER FOR THE COST OF CHANGING THE DRAWINGS AND DETAILS SHOULD THIS BE REQUIRE BY THE BUILDING DEPARTMENT OR SHOULD THE ENGINEER DECIDE IT IS NECESSARY.
- C. THE SUBSTITUTION SHALL BE PASSED ON TO THE OWNER SHOULD THE SUBSTITUTION BE APPROVED.
18. THE CONTRACTOR SHALL NOTIFY THE ARCHITECT AND STRUCTURAL ENGINEER WHEN A CONFLICT OR DISCREPANCY OCCURS BETWEEN THE STRUCTURAL DRAWINGS AND ANY OTHER PORTION OF THE CONTRACT DOCUMENTS OR EXISTING FIELD CONDITIONS. SUCH NOTIFICATION SHALL BE GIVEN IN DUE TIME SO AS NOT TO INTERFERE WITH THE CONSTRUCTION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE STRUCTURAL DRAWINGS AND SPECIFICATIONS THE MORE RESTRICTIVE CONDITION SHALL TAKE PRECEDENCE UNLESS WRITTEN APPROVAL, HAS BEEN GIVEN FOR THE MORE RESTRICTIVE CONDITION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL DIMENSIONS WITH ARCHITECTURAL PRIORITY TO COMMENCING ANY WORK.

FOUNDATION NOTES:

1. DESIGN: FOUNDATION DESIGNED IN CONFORMANCE WITH THE RECOMMENDATIONS BY SPEEDE AND ASSOCIATES
REPORT NO.S: 2100256A
DATED: MARCH 05, 2021
DESIGNED BY: KEITH R. GRAVEL
PHONE # (602) 987-6381
ADDRESS: 3331 EAST WOOD STREET, PHOENIX, AZ 85040
2. FOOTINGS: ALL FOOTINGS SHALL EXTEND A MINIMUM DEPTH BELOW FINISHED OR NATURAL GRADE INTO ACCEPTABLE GEOLOGICAL MATERIAL AS FOLLOWS UNLESS NOTED OTHERWISE:
 - 2.1. EXTERIOR FOOTINGS = 24"
 - 2.2. INTERIOR FOOTINGS = 24"
 - 2.3. FOUNDATION WALL FOOTINGS = 24"
3. FOUNDATION DESIGN SOIL VALUES
 - 3.1. ALLOWABLE BEARING PRESSURE = 1500 PSF
 - 3.2. ALLOWABLE PASSIVE PRESSURE = 250 PCF
 - 3.3. COEFFICIENT OF FRICTION = 0.25
 - 3.1. ALLOWABLE BEARING PRESSURE MAY BE INCREASED $\frac{1}{3}$ FOR WIND OR SEISMIC
 - 3.2. ALLOWABLE PASSIVE AND FRICTION MAY BE COMBINED 100% TO RESIST SLIDING
4. BACKFILL: PARTIAL BACKFILLING OF RETAINING WALLS MAY COMMENCE ONLY WITH WRITTEN DIRECTION FROM THE ENGINEER OF RECORD (EOR).

CONCRETE NOTES

1. COMPRESSIVE STRENGTH: THE MINIMUM ULTIMATE COMPRESSIVE STRENGTH OF ALL CONCRETE SHALL BE 4500 PSI MIN AT 28 DAYS, UNLESS NOTED OTHERWISE. REFER TO PLANS AND TABLE BELOW FOR THE DESIGN STRENGTH OF CONCRETE FOR SPECIFIC STRUCTURAL ELEMENTS. DESIGN OF MIXES SHALL BE BY AN APPROVED TESTING LABORATORY AND SIGNED BY A REGISTERED ENGINEER.

1.1. SLAB ON GRADE	$f_c = 4500$ PSI MIN
1.2. CONVENTIONAL FOOTINGS	$f_c = 4500$ PSI MIN
1.3. GRADE BEAMS	$f_c = 4500$ PSI MIN
1.4. ALL CONCRETE IN CONTACT WITH SOIL	$f_c = 4500$ PSI MIN
2. WEIGHT: ALL CONCRETE SHALL BE "NORMAL WEIGHT" UNLESS NOTED OTHERWISE.
3. CONCRETE SHALL HAVE A MAXIMUM WATER-CEMENTITIOUS MATERIAL RATIO, BY WEIGHT, OF 0.45
4. CEMENT: CEMENT SHALL CONFORM TO THE ASTM C150-09 TYPE I V.
5. AGGREGATE: AGGREGATE SHALL CONFORM TO ASTM C33-08.
6. SPECIAL INSPECTIONS: SEE THE SPECIAL INSPECTION TABLE AND THE SPECIAL INSPECTION NOTE SHEET FOR ANY INSPECTIONS REQUIRED TO BE MADE.
7. CONCRETE PLACEMENT AND QUALITY: SHALL CONFORM TO APPLICABLE RECOMMENDATIONS OF ACI SP-15. A COPY OF SP-15 SHALL BE AVAILABLE AT CONSTRUCTION SITE DURING THE PROJECT.
8. DEBRIS: REMOVE ALL DEBRIS FROM FORMS BEFORE PLACING OF CONCRETE.
9. DOWELING: ALL WALLS AND COLUMNS SHALL BE DOWELED INTO FOOTINGS, WALLS, BEAMS, OR SLABS AS SHOWN OR NOTED ON THE DRAWINGS.
10. SPLICES: VERTICAL WALL BEAMS SHALL BE SPLICED AT OR NEAR FLOOR SLABS. SPLICE BARS IN SPANDRELS, WALLS, BEAMS, GRADE BEAMS, ETC., UNLESS NOTED OTHERWISE AS FOLLOWS: TOP BARS AT MID-POINT OF SPAN, BOTTOM BARS AT THE SUPPORT. ALL REINFORCEMENT SHALL BE SECURELY WIRED AND POSITIVELY SUPPORTED ABOVE THE GROUND AND AWAY FROM FORMS AS SHOWN OR NOTED.
11. INSERTS: ALL ITEMS TO BE CAST IN CONCRETE SUCH AS REINFORCING, DOWELS, BOLTS, ANCHORS, PIPES, SLEEVES, ETC., SHALL BE SECURE AND POSITIONED BEFORE PLACING OF CONCRETE.
12. CONDUIT AND PIPES: CONDUIT AND PIPES SHALL NOT BE EMBEDDED IN STRUCTURAL CONCRETE EXCEPT WHERE SPECIFICALLY APPROVED BY THE ENGINEER OF RECORD. REINBURD CONDUIT AND PIPE SIZE SHALL BE 1/3 OF THE SLAB WALL THICKNESS AND LOCATED AT ITS MID DEPTH. MINIMUM SPACING SHALL BE 3 TIMES THE CONDUIT/PIPE DIAMETER. CONDUIT AND PIPES SHALL NOT IMPAIR THE STRENGTH OF THE MEMBER. CONDUIT SHALL BE SECURELY SUPPORTED ABOVE THE CONCRETE AREA. SHALL NOT DISPLACE MORE THAN 4% OF THE CROSS SECTION AREA.
13. CONSTRUCTION JOINTS: SHALL HAVE ENTIRE SURFACE REMOVED TO EXPOSE CLEAN AGGREGATE SOLIDLY EMBEDDED. THE CONTRACTOR SHALL OBTAIN THE APPROVAL OF CONSTRUCTION JOINT LOCATION IN ALL SLABS, BEAMS, AND WALLS.
14. NON-SHRINK CEMENT GROUT SHALL HAVE A MINIMUM 28 DAY COMPRESSIVE STRENGTH OF 9000 PSI USE "SIKA GROUT 212" OR MASTERFLOW 828

REINFORCING STEEL NOTES

1. GRADE: ALL REINFORCING STEEL SHALL BE DEFORMED BARS WHICH SHALL CONFORM TO THE STANDARD SPECIFICATIONS OF ASTM A615 GRADE 60. UNO. ALL REINFORCEMENT TO BE WELDED SHALL BE ASTM A706 GRADE 60. UNO. ALL REINFORCEMENT THAT IS PART OF THE SEISMIC FORCE RESISTING SYSTEM SHALL BE ASTM A706 GRADE 60.
2. WELDED WIRE MESH: ELECTRIC WELDED WIRE MESH SHALL CONFORM TO ASTM A-185. SIZE SHALL BE AS SHOWN ON THE DRAWINGS. MINIMUM LAPS TO BE 12".
3. MINIMUM LAP: SEE LAPS/SPACE DETAILS.
4. MINIMUM COVER: REINFORCING STEEL TO HAVE THE FOLLOWING MINIMUM COVER:

A. CONCRETE AGAINST EARTH (NOT FORMED)	3"
B. CONCRETE EXPOSED TO EARTH OR WEATHER (FORMED OR TROWELED)	@ CL OF SLAB
#6 - #18 BARS	2"
#5 AND SMALLER	1"
C. SLAB-ON-GRADE	@ CL OF SLAB
D. WALLS	1 1/2"
E. CONCRETE SLAB (FORMED)	1"
F. STRUCTURAL SLAB AND WALLS	1"
G. BEAMS AND COLUMNS	1"
5. (PRIMARY REINFORCEMENT, TIES, STIRRUPS, SPIRALS) 1 1/2"
6. WELDING: LOW-HYDROGEN WELDING RODS SHALL BE USED FOR ALL WELDING TO REINFORCING BARS, BUT ONLY WHERE SHOWN OR NOTED BY THE STRUCTURAL ENGINEER.
7. WELDING: GRADE 60 REBAR SHALL BE PREHEATED, WHEN WELDING, AS PRESCRIBED BY AWS D1.4 FOR VARIOUS SIZE BARS. REBAR SHALL BE ASTM A 706 GRADE 60.
8. DOWELING: DOWELS SHALL BE PROVIDED AT CONNECTION JOINTS AND SHALL BE THE SAME SIZE AND SPACING AS DETAILED OR #3 @ 12" O.C. X 3'-0" (MINIMUM).
9. TOLERANCE FOR REBAR PLACEMENT: TOLERANCE FOR LONGITUDINAL LOCATION OF BENDS AND ENDS OF REINFORCEMENT SHALL BE PLUS OR MINUS 2 INCHES EXCEPT AT DISCONTINUOUS ENDS OF MEMBERS WHERE TOLERANCES SHALL BE PLUS OR MINUS 1/2 INCH.
10. TEMPERATURE REINFORCEMENT: SHALL HAVE A LAP OF 48 BAR DIAMETERS BUT NOT LESS THAN 24 INCHES, AND THE SPLICE IN ADJACENT BARS SHALL BE NOT LESS THAN 5 FEET APART.

STRUCTURAL STEEL NOTES:

- STEEL FRAMES: ALL STRUCTURAL STEEL FOR FRAMES SHALL CONFORM TO ASTM A992 GR. 50 PER AISI. ALL WELDS FOR STEEL FRAMES SHALL BE OF A NOTCH TOUGH WELD METAL (AWS D10.10F/D10.10F + D10.10F/D10.10F F) AND E70XX (70ksi) ELECTRODES. WELDS SHALL BE MADE BY THE SHOWN WELDING OR EQUIVALENT ACCORDING TO (AWS) REQUIREMENTS, UNLESS OTHERWISE SPECIFIED IN THESE DRAWINGS. COMPLETE JOINT PENETRATION WELDS SHALL HAVE THE BACKUP BARS REMOVED. THE EXPOSED WELD BACK GROUTED (GROUND TO SOUND BRIGHT METAL), AND FINALLY REINFORCED WITH A MINIMUM 3/8" FILLET WELD TO COMPLY WITH CURRENT STANDARDS.
1. SHOP DRAWINGS: SHOP DRAWINGS SHALL BE SUBMITTED TO THE ARCHITECT FOR REVIEW AND COMMENT PRIOR TO FABRICATION.
2. DETAILING: ALL CONNECTIONS AND DETAILING PRACTICES SHALL CONFORM TO THE 14TH EDITION OF A.I.S.C. SPECIFICATIONS.
3. CERTIFICATION: AT COMPLETION OF FABRICATION, THE APPROVED FABRICATOR SHALL SUBMIT A CERTIFICATE OF COMPLIANCE TO THE BUILDING OFFICIAL STATING THAT THE WORK WAS PERFORMED IN ACCORDANCE WITH THE APPROVED CONSTRUCTION DOCUMENTS.

SPECIAL INSPECTION NOTES:

1. GENERAL: IN ADDITION TO THE INSPECTIONS REQUIRED BY SECTION 110 OF THE 2019 CBC, THE OWNER SHALL EMPLOY A SPECIAL INSPECTOR DURING CONSTRUCTION ON THE FOLLOWING TYPES OF WORK: ALL SPECIAL INSPECTIONS SHALL BE PERFORMED IN ACCORDANCE WITH CHAPTER 17 OF THE 2024 IBC.
2. CONCRETE: SEE SPECIAL INSPECTION SHEET
3. STRUCTURAL STEEL: SEE SPECIAL INSPECTION SHEET
4. EPOXY ANCHORS: REFER TO ICC-ES REPORTS
5. SELECTION OF THE SPECIAL INSPECTOR: THE OWNER SHALL SUBMIT TO THE ARCHITECT A LIST OF 3 FIRMS CHOSEN TO PERFORM THE SPECIAL INSPECTION DUTIES. THE SPECIAL INSPECTION FIRM SHALL HAVE AT LEAST 5 YEARS OF EXPERIENCE IN THE WORK TO BE INSPECTED. THE ARCHITECT SHALL RECOMMEND A FIRM FROM THOSE SUBMITTED.
6. FIELD INSPECTOR: ALL FIELD INSPECTORS SHALL HAVE A MINIMUM OF 2 YEARS EXPERIENCE IN THE SPECIFIC CONSTRUCTION BEING INSPECTED

STRUCTURAL STEEL AND MISCELLANEOUS METAL (ALL OTHER STEEL) WELDING

1. ALL WELDING AND WELDED JOINTS SHALL BE IN STRICT CONFORMANCE WITH THE LATEST EDITION OF AWS D 1 AND THE CALIFORNIA BUILDING CODE WITH ALL APPLICABLE AMENDMENTS. ALL NON PRE-QUALIFIED WELDED JOISTS SHALL BE PRE-QUALIFIED BY TEST. THE TEST RECORDS SHALL BE IN ACCORDANCE WITH THE LATEST EDITION OF AWS D1.1
2. WELDING OF SHEET METAL AND METAL STUDS SHALL BE IN ACCORDANCE WITH AWS D1.3
3. A WRITTEN "WELDING PROCEDURE SPECIFICATION" (WPS) SHALL BE DEVELOPED BY THE FABRICATOR/ERECTOR, AND REVIEWED BY THE OWNER'S REPRESENTATIVE AND WELDING DEPARTMENT. THE WPS SHALL CONTAIN ALL THE NECESSARY INFORMATION REQUIRED BY THE CODE, THE SPECIFICATIONS, AND ANY OTHER INFORMATION NECESSARY TO PRODUCE WELDS THAT ARE IN COMPLIANCE WITH THESE REQUIREMENTS. THE WPS SHALL INCLUDE THE WELDING PARAMETERS RECOMMENDED BY THE ELECTRODE MANUFACTURE. ALL WELDERS AND INSPECTORS SHALL ADHERE TO THE WPS AND SHALL RETAIN A COPY.
4. ALL WELDS SHALL HAVE A FLIPER METAL, WITH CHARPY V NOTCH TOUGHNESS OF 20 FT-LBS AVERAGE AT MINUS TWENTY DEGREES FAHRENHEIT AND 40 FT-LBS AT SEVEN DEGREES FAHRENHEIT. CERTIFY CONFORMANCE TO CHARPY V NOTCH TOUGHNESS REQUIREMENTS WITH TEST BY AN INDEPENDENT TESTING LABORATORY.
5. WELD LENGTHS CALLED FOR ON THE PLANS ARE THE NET EFFECTIVE LENGTH REQUIRED. WELD SIZE SHALL BE AISC MINIMUMS UNLESS A LARGER SIZE IS NOTED. WHERE LENGTH OF WELD IS NOT SHOWN, IT SHALL BE FULL LENGTH OF JOINT. ALL BUTT AND GROOVE WELDS SHALL BE FULL PENETRATION, UNLESS NOTED OTHERWISE.
6. ALL WELDING ELECTRODES AND ELECTRODE FLUX COMBINATIONS (FLIPER METAL) SHALL BE MINIMUM TO KSI, UNO, AND SHALL MEET THE REQUIREMENTS PER AISC SEISMIC PROVISIONS.
7. GMAW AND FCAG-G WELDING PROCESSES SHALL NOT BE PERMITTED WHEN WIND SPEED EXCEEDS 5 MPH.
8. WHERE FIELD WELDING IS NOTED, THE DESIGNATION IS GIVEN AS A SUGGESTED CONSTRUCTION PROCEDURE ONLY. THE CONTRACTOR IS SOLELY RESPONSIBLE FOR IDENTIFY THE METHOD OF FABRICATION.
9. ALL SHOP WELDS SHALL BE PERFORMED BY A FABRICATOR LICENSED BY THE LOCAL JURISDICTION.
10. ALL WELDERS SHALL BE QUALIFIED FOR THE WORK THEY WILL BE PERFORMING SHALL HAVE CURRENTLY VALID CERTIFICATIONS ISSUED BY AWS AND THE GOVERNING JURISDICTION.

SIMPSON STRONG-TIE SET-3G EPOXY ADHESIVE ANCHORS

1. ANCHOR MANUFACTURER: WHERE ADHESIVE ANCHORS ARE CALLED FOR ON THESE PLANS, THE CONTRACTOR SHALL USE THE "SIMPSON STRONG-TIE SET-3G EPOXY ADHESIVE ANCHOR SYSTEM" BY SIMPSON STRONG-TIE COMPANY INC. INSTALLED PER REQUIREMENTS OF ICC-ES REPORT NO. ESR-4507, DATED APRIL 2025.
2. ADHESIVE REQUIREMENT: THE ADHESIVE USED WITH THESE ANCHORS SHALL BE "SIMPSON SET-3G EPOXY ADHESIVE", INSTALLATION OF WHICH SHALL PER THE ABOVE REFERENCED ICC-ES REPORT AND BE UNDER THE CONTINUOUS OBSERVATION OF A SPECIAL INSPECTOR PER THE ICC-ES REPORT.
3. ANCHOR RODS: ANCHOR RODS SHALL BE CONTINUOUSLY THREADED STEEL RODS CONFORMING TO ASTM A 193, GRADE 87.

ABBREVIATIONS

AB	AT	L	ANGLE
AB	ANCHOR BOLT	LA	LATERAL
ABV	ABOVE	LBS. #	POUNDS
ACC	AMERICAN CONCRETE INSTITUTE	LINE FEET	(FOOT)
ADD	ADDITIONAL	LV	LOAD
ADVL	AMERICAN INSTITUTE OF STEEL	LLH	LONG LEG HORIZONTAL
ACI	CONSTRUCTION	LLV	LONG LEG VERTICAL
AISC	AMERICAN PLYWOOD ASSOCIATION	LONG	LONGITUDINAL
APPROX	APPROXIMATE	LVL	LUMBER STRAND
ARCH	ARCHITECT	LSL	MICROLAM
ARCHL	ARCHITECTURAL		
ASTM	AMERICAN SOCIETY FOR TESTING & MATERIALS	MATL	MATERIAL
AWS	AMERICAN WELDING SOCIETY	MB	MACHINE BOLT
		MECHL	MECHANICAL
BI	BOTTOM OF	MEMB	MEMBRANE
BL	BOUNDARY ELEMENT	MANF	MANUFACTURER
BLW	BELOW	MIN	MINIMUM
BLK	BUILDING	MISC	MISCELLANEOUS
BLKG	BLOCKING	MTL	METAL
BM	BEAM		
BN	BOUNDARY NAILING	(N)	NEW
BO	BOTTOM OF (FOOTING, BEAM, ETC)	N/A	NON APPLICABLE
BTM	BOTTOM	NIC	NOT IN CONTRACT
BTWN	BETWEEN	NO	NUMBER
		NOM	NOMINAL
		NTE	NOT TO EXCEED
CALCS	CALCULATIONS	NTS	NOT TO SCALE
CANT	CANTILEVER		
CF	CUBIC FOOT	O/	OVER
CFS	COLD FORMED STEEL	OC	ON CENTER
CON	CONCRETE	OD	OUTSIDE DIAMETER
CL	CENTERLINE OR CLEAR	OF / OSF	OUTSIDE FACE
CLR	CLEAR	OPNG	OPENING
CMU	CONCRETE MASONRY UNIT	OPP	OPPOSITE
COL	COLUMN	OPTL	OPTIONAL
CONC	CONCRETE	OWNJ	OPEN WEB WOOD JOIST
CONN	CONNECTION		
CONSTR	CONSTRUCTION	PA	POST ABOVE
CONT	CONTINUOUS	PB	POST BELOW
CONTR	CONTRACTOR	PAR	PARALLEL
COORD	COORDINATE	PEN	PENETRATION
CTR	CENTERED	PERP	PERPENDICULAR
CTSK	COUNTERSINK	PL	PLATE
		PLT	POUNDS PER LINEAR FOOT
D	PENNY	PLY	PLYWOOD
D	DEPTH	PRELIM	PRELIMINARY
DBA	DEFORMED BAR ANCHOR	PPS	POUNDS PER SQUARE FOOT
DBL	DOUBLE	PSI	POUNDS PER SQUARE INCH
DF	DOUGLAS FIR	PSL	PARALLEL STRAND LUMBER
DI	DIAMETER	PT	PRESSURE TREATED
DIAPH	DIAPHRAGM		
DIM	DIMENSION	QTY	QUANTITY
DKG	DECKING		
DL	DEAD LOAD	REF	REFERENCE
DTL	DETAIL	REINF	REINFORCEMENT
DWG	DRAWING	REQD	REQUIRED
DWL	DOWEL	RET	REINFORCING
		REV	REVISION
		REV	REVISION
(E)	EXISTING	SCHED	SCHEDULE
EA	EACH	SF	SQUARE FEET (FOOT)
EE	EACH END	SFRS	SEISMIC FORCE RESISTING SYSTEM
ELEV	ELEVATION	SHT	SHEET
ENG	ENGINEER	SHTG	SHEATHING
EQ	EQUAL	SIM	SIMILAR
EQUIP	EQUIPMENT	SIN	SILL NAIL
EQS	EACH SIDE	SOQ	SLAB ON GRADE
EW	EACH WAY	SPEC	SPECIFICATION
		SQ	SQUARE
F	FACE OF	S.S.	STAINLESS STEEL
FB	FLAT BAR	SS	STRUCTURAL SELECT
FDN	FOUNDATION	STD	STANDARD
FF	FINISHED FLOOR	STL	STEEL
FLG	FINISHED GRADE	STR	STRUCTURAL
FLR	FLOOR	STYL	SQUARE YARD
FN	FIELD NAIL		
FOC	FACE OF CONCRETE	T	TOP
FOS	FACE OF STEEL	T & B	TOP & BOTTOM
FRMG	FRAMING	T & G	TONGUE & GROOVE
FT	FEET (FOOT)	TOP	TOP OF
FTG	FOOTING	THRD	THREADED
		THKN	THICKENED
GA / ga	GAUGE	TL	TOTAL LOAD
GLV	GALVANIZED	TN	TOE NAIL
GEN	GENERAL	TOF	TOP OF FOOTING / FOUNDATION
GR	GRADE	TOSH	TOP OF SHEATHING
HD	HOLDOWN	TOSI	TOP OF SLAB
HDR	HEADER	TOSI	TOP OF STEEL
HGR	HANGER	TRANV	TRANSVERSE
HSS	HOLLOW STRUCTURAL SECTION	TY	TYPICAL
HGT	HEIGHT	UNO	UNLESS NOTED OTHERWISE
IBC	INTERNATIONAL BUILDING CODE	VAR	VARIES
ID	INSIDE DIAMETER	VERT	VERTICAL
IF / ISF	INSIDE FACE	VF	VERIFY IN FIELD
IN	INCHES	VL	VERSA-LAM
INFO	INFORMATION		
INSP	INSPECTION	W/	WITH
INT	INTERIOR	WO	WITHOUT
INTER	INTERMEDIATE	WD	WOOD
		WF	WIDE FLANGE
		WS	WOOD SCREWS
K	KIPS		
KSF	KIPS PER SQUARE FOOT		
KSI	KIPS PER SQUARE INCH		
KO	KNOCK OUT		
KP	KING POST		

BASIS OF DESIGN

- | | | | |
|----------------------|--|-----------|----------------------------|
| 1. | BUILDING CODE: 2024 EDITION OF THE INTERNATIONAL BUILDING CODE (IBC) | | |
| BUILDING LIVE LOADS: | | | |
| | ROOF | | 20 PSF** |
| | FLOOR | | 100 PSF |
| | **REDUCIBLE DEPENDING UPON TRIBUTARY AREA, AND SLOPE | | |
| 3. | WIND LOADS: | | |
| | ULTIMATE WIND SPEED | V_{ULT} | = 101 MPH |
| | NOMINAL DESIGN WIND SPEED: | | |
| | $V_{ASD} = V_{ULT} \cdot \text{"SORT"}(0.6)$ | | = 78 MPH |
| | RISK CATEGORY | | II |
| | EXPOSURE | | D |
| | INTERNAL PRESSURE COEFFICIENT (GCp1) | | = ±0.18 |
| 4. | RISK CATEGORY: | | |
| | RISK CATEGORY | | II |
| | SEISMIC IMPORTANCE FACTOR | | 1.0 |
| 5. | SEISMIC LOADS: | | |
| | SITE CLASS | | D |
| | SEISMIC DESIGN CATEGORY | | D |
| | RESPONSE MODIFICATION FACTORS | | D |
| | | R = | 4.0 (CONCRETE SHEAR WALLS) |
| | MAPPED SPECTRAL RESPONSE ACCELERATIONS | | |
| | | S_s | = 0.17 |
| | | S_1 | = 0.059 |
| 6. | SPECTRAL RESPONSE COEFFICIENTS | | |
| | | S_{sw} | = 0.19 |
| | | S_{d1} | = 0.11 |

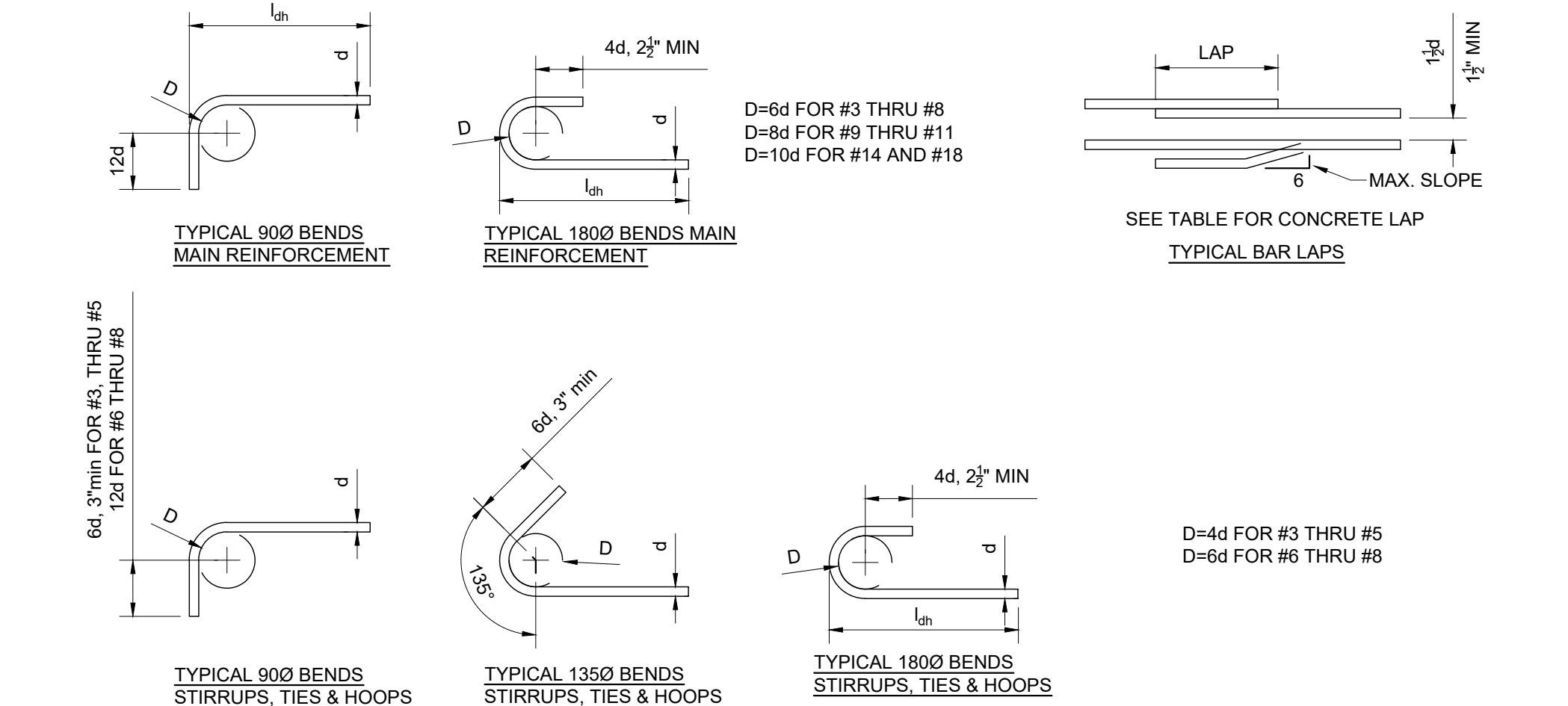
STRUCTURAL GENERAL NOTES

HACE, INC

TENANT IMPROVEMENT WEST 202 LOGISTICS

DATE	2026-02-10
SCALE	AS SHOWN
PROJECT MANAGER	MEB
CHECKED BY	MEB
DRAWN BY	NMN
JOB NO.	26011
SHEET	

S001



STRAIGHT BAR DEVELOPMENT, L _d								
F _y =60 ksi NORMAL-WEIGHT CONCRETE								
BAR SIZE	f _c =2500 psi		f _c =3000 psi		f _c =4000 psi		f _c =5000 psi	
	TOP	OTHER	TOP	OTHER	TOP	OTHER	TOP	OTHER
#3	20"	15"	18"	14"	16"	12"	14"	12"
#4	32"	24"	29"	22"	25"	19"	23"	17"
#5	45"	35"	42"	32"	36"	28"	32"	25"
#6	61"	47"	55"	43"	48"	37"	43"	33"
#7	96"	74"	88"	68"	76"	59"	68"	52"
#8	117"	90"	107"	83"	93"	72"	83"	64"
#9	132"	102"	121"	93"	105"	81"	94"	72"
#10	149"	115"	136"	105"	118"	91"	106"	81"
#11	165"	127"	151"	116"	131"	101"	117"	90"
#14	199"	153"	181"	140"	157"	121"	141"	108"
#18	265"	204"	242"	186"	209"	161"	187"	144"

LAP SPLICE SCHEDULE, CLASS B								
Fy=60 ksi NORMAL-WEIGHT CONCRETE, 1.3Ld								
BAR SIZE	fc=2500 psi		fc=3000 psi		fc=4000 psi		fc=5000 psi	
	TOP	OTHER	TOP	OTHER	TOP	OTHER	TOP	OTHER
#3	25"	20"	23"	18"	20"	16"	18"	16"
#4	41"	32"	38"	29"	33"	25"	29"	23"
#5	59"	45"	54"	42"	47"	36"	42"	32"
#6	79"	61"	72"	55"	62"	48"	56"	43"
#7	125"	96"	114"	88"	99"	76"	88"	68"
#8	153"	117"	139"	107"	121"	93"	108"	83"
#9	172"	132"	157"	121"	136"	105"	122"	94"
#10	194"	149"	177"	136"	153"	118"	137"	106"
#11	215"	165"	198"	151"	170"	131"	152"	117"

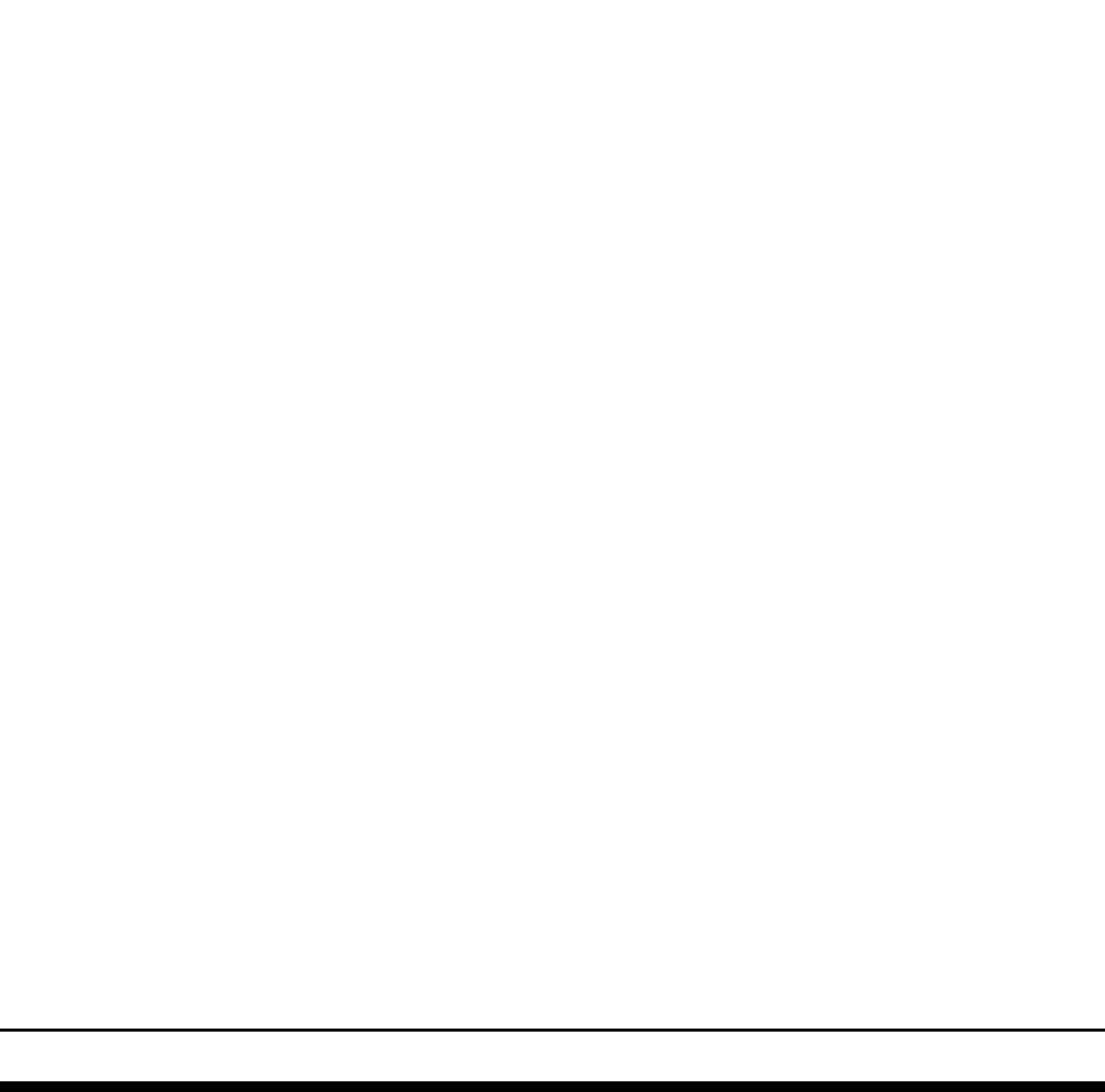
STANDARD HOOK DEVELOPMENT, l _h				
F _y =60 ksi NORMAL-WEIGHT CONCRETE				
BAR SIZE	f _c =2500psi	f _c =3000psi	f _c =4000psi	f _c =5000psi
#3	9"	9"	8"	7"
#4	12"	11"	10"	9"
#5	15"	14"	12"	11"
#6	18"	17"	15"	13"
#7	21"	20"	17"	15"
#8	24"	22"	19"	17"
#9	28"	25"	22"	20"
#10	31"	28"	25"	22"
#11	34"	31"	27"	24"
#14	41"	38"	33"	29"
#18	55"	50"	43"	39"

- SEISMIC DEVELOPMENT AND LAP LENGTHS:
- MINIMUM SEISMIC DEVELOPMENT AND LAP LOCATIONS
 - SPECIAL CONCRETE SHEAR WALLS
 - LAP OF VERTICAL REINFORCEMENT AT BASE OF WALL
 - DEVELOPMENT OF REINFORCEMENT IN FOUNDATION ELEMENT
 - DEVELOPMENT OF DIAGONAL REBAR OF COUPLER BEAMS INTO WALLS
 - SPECIAL CONCRETE MOMENT FRAMES
 - DEVELOPMENT OF LONGITUDINAL BEAM REINFORCEMENT IN CONFINED CORE OF COLUMN. ANY PORTION OF STRAIGHT BAR DEVELOPMENT OUTSIDE THE CONFINED CORE SHALL BE INCREASED BY A FACTOR OF 1.6
 - COLUMN LONGITUDINAL REBAR DEVELOPMENT INTO FOUNDATION ELEMENTS.
 - SEE PLAN AND DETAILS FOR ADDITIONAL SEISMIC DEVELOPMENT AND LAP LOCATIONS
- NOTES:
- TOP BARS ARE DEFINED AS HORIZONTAL REINFORCEMENT WHERE MORE THAN 12" OF FRESH CONCRETE IS PLACED BELOW THE BARS BEING DEVELOPED OR SPLICED
 - VALUES IN TABLES ABOVE ARE FOR NORMAL WEIGHT CONCRETE ONLY. INCREASE LENGTHS BY 33.3% (MULTIPLY BY 1.33) FOR LIGHTWEIGHT CONCRETE
 - #14 AND #18 REQUIRE WELDED SPLICE OR MECHANICAL SPLICE
 - FOR F_c GREATER THAN 5000 PSI, USE VALUES FOR F_c = 5000 PSI.
 - FOR F_c = 4500 PSI, USE VALUES FOR F_c = 4000 PSI.

CONCRETE REBAR DEVELOPMENT, LAP SPLICES AND BENDS

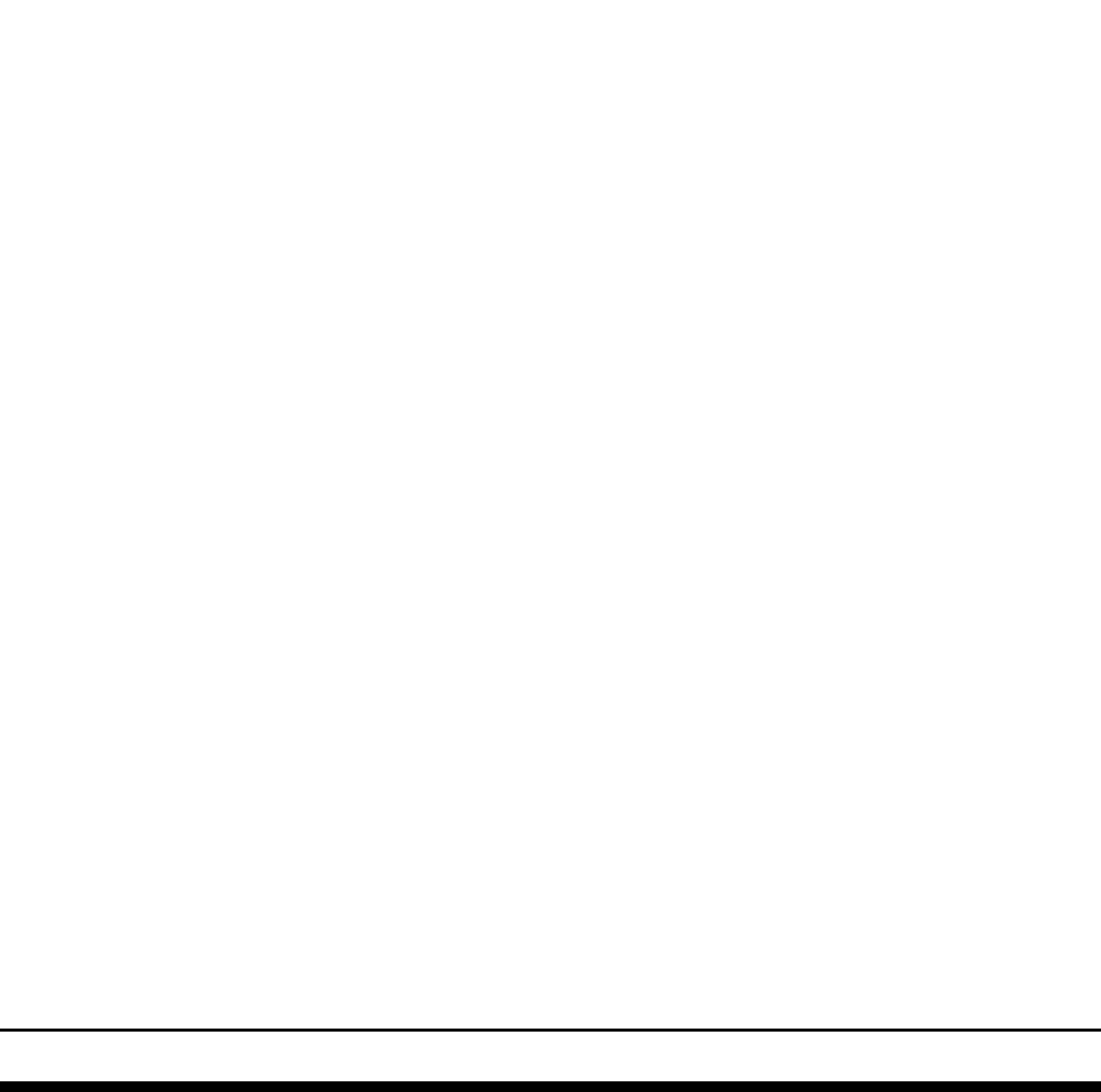
11 DETAIL

scale 3/4"=1'-0"



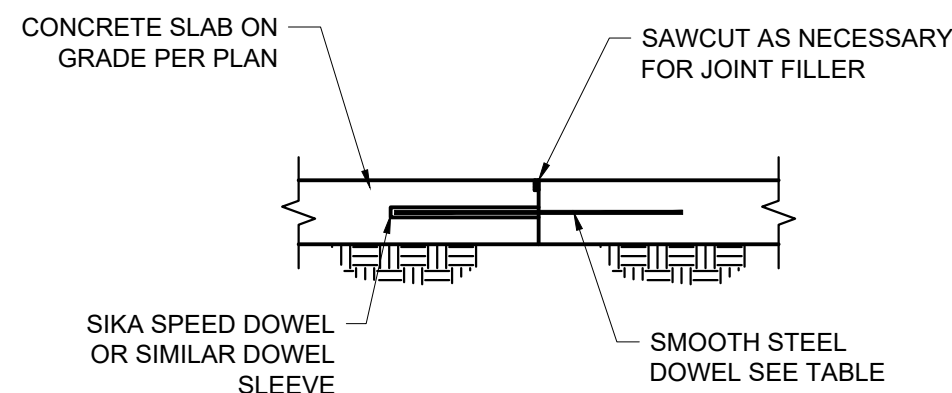
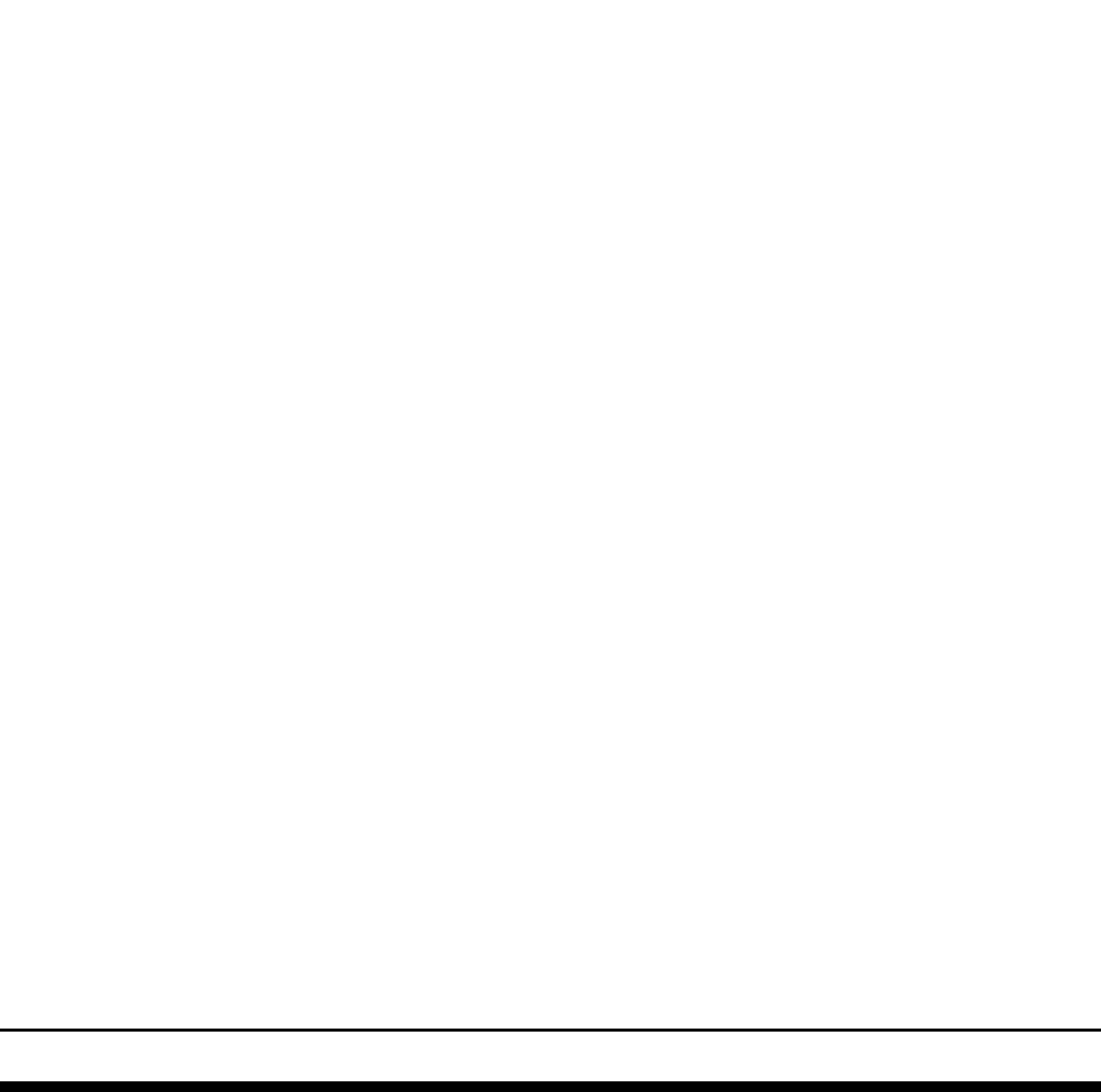
13 DETAIL

scale 3/4"=1'-0"



14 DETAIL

scale 3/4"=1'-0"

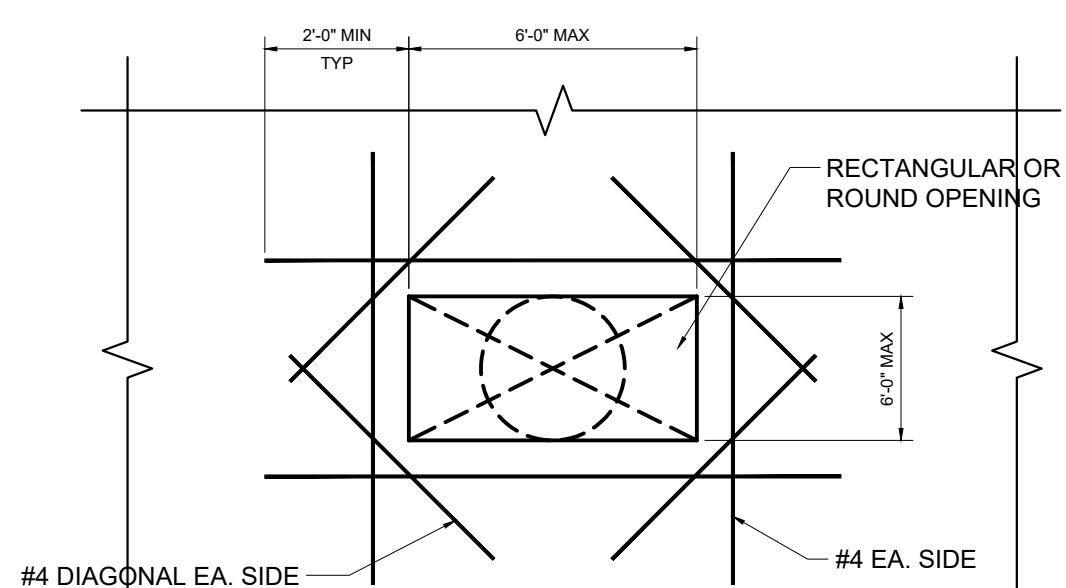


DOWEL CONSTRUCTION JOINT

- NOTES:
- INSTALL DOWEL SLEEVE PER MANUFACTURE INSTALLATION INSTRUCTIONS.
 - INSTALL DOWEL INTO SLEEVE PER MANUFACTURE INSTRUCTIONS.
 - 2" RADIUS EACH SIDE OF JOINT AS REQUIRED TO PROPERLY LOCATE SAW CUTS

DOWEL SIZE AND SPACING FOR CONSTRUCTION JOINT		
SLAB THICKNESS INCHES	DOWEL SIZE	SPACING INCHES ON CENTER
<= 6"	6.75" DIA X 18"	18"
>6" to 8"	1.0" DIA X 18"	12"
>8"	1.25" DIA X 18"	12"

- NOTES:
- DOWELS SHALL BE A36 OR GREATER STRENGTH STEEL

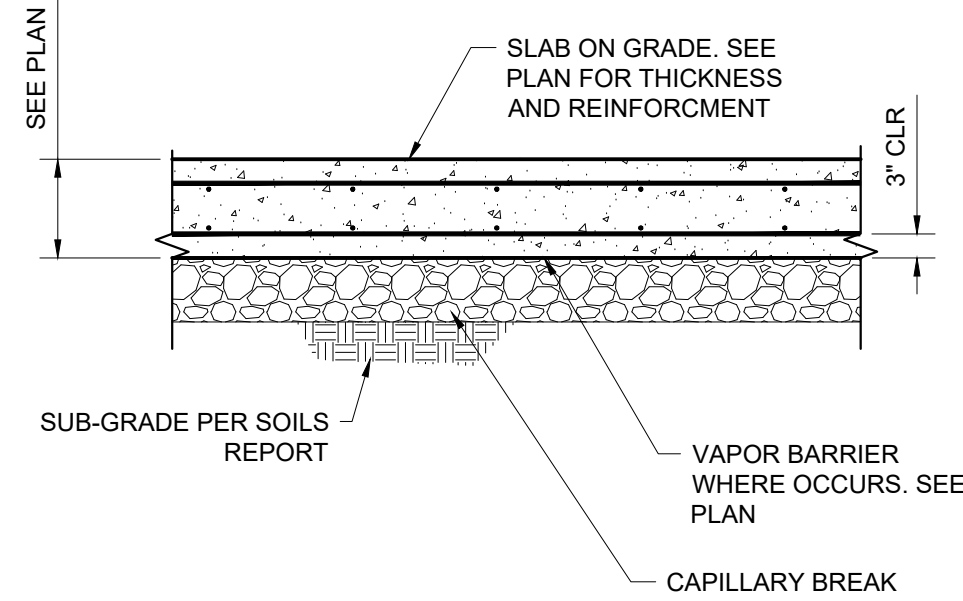


- NOTES:
- REINFORCEMENT NOT REQ'D FOR OPENINGS LESS THAN 24"x24"
 - FOR OPENINGS LARGER THAN 8'1"x8'1", USE TURNED DOWN SLAB ON GRADE EDGE DETAIL

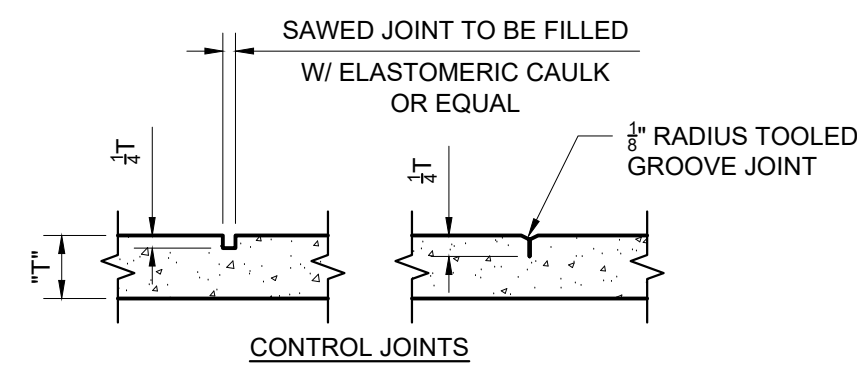
scale 3/4"=1'-0"

5 DETAIL

scale 3/4"=1'-0"



- NOTES:
- SEE SOILS REPORT FOR CAPILLARY BREAK INFORMATION. MINIMUM REQUIREMENT IS 4" OF AGGREGATE.
 - SEE ARCHITECTURAL DRAWINGS FOR VAPOR BARRIER. 15MIL MIN MINIMUM REQUIREMENT IS A 15 MIL VAPOR BARRIER.

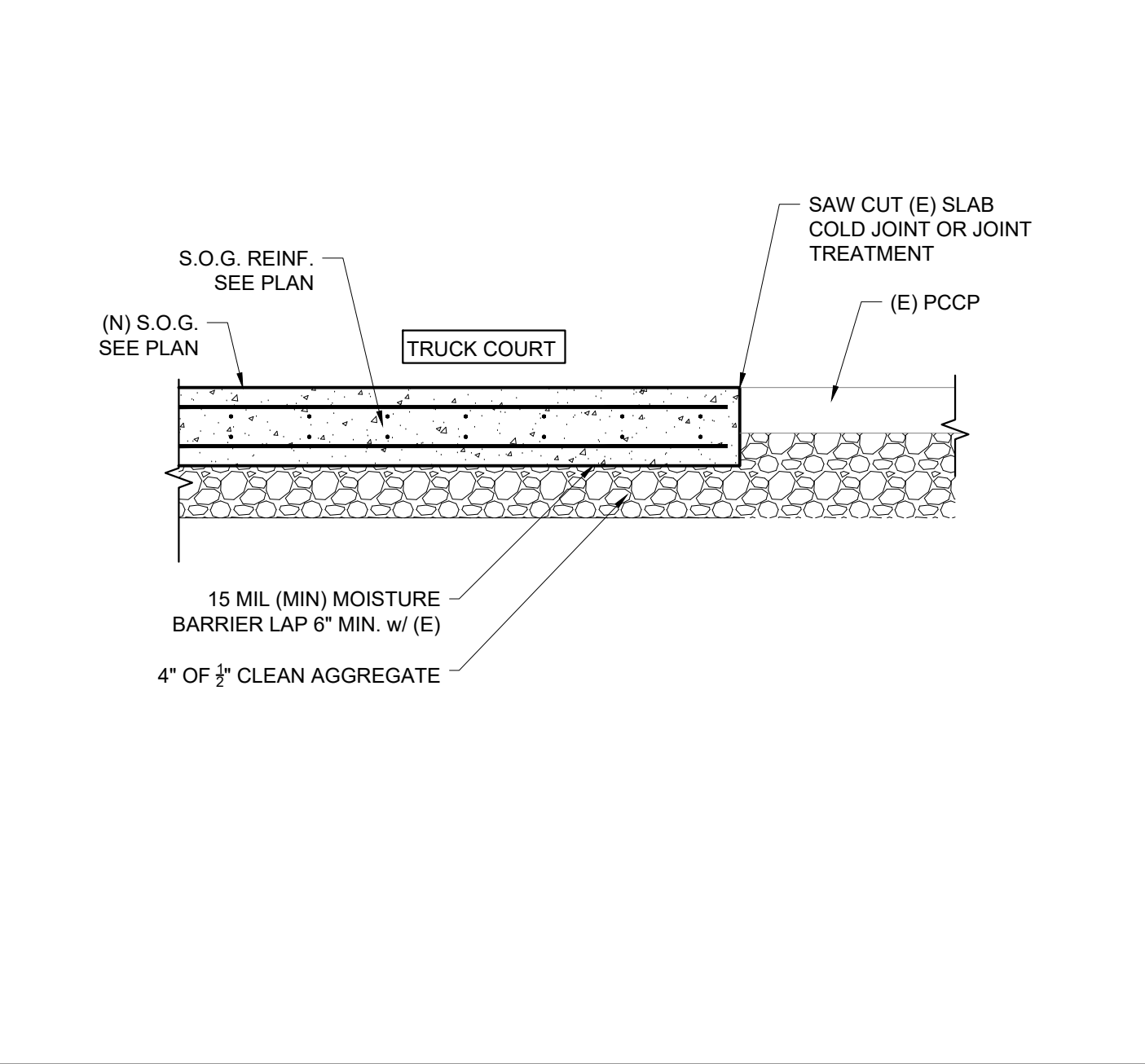


- NOTE:
- SLAB JOINTS SHALL BE INSTALLED AT INTERVALS NOT EXCEEDING 12'-0" IN BOTH DIRECTIONS (PREFERABLY A SQUARE PATTERN BUT NOT TO EXCEED 1:1/2)

SLAB ON GRADE CRACK CONTROL JOINT

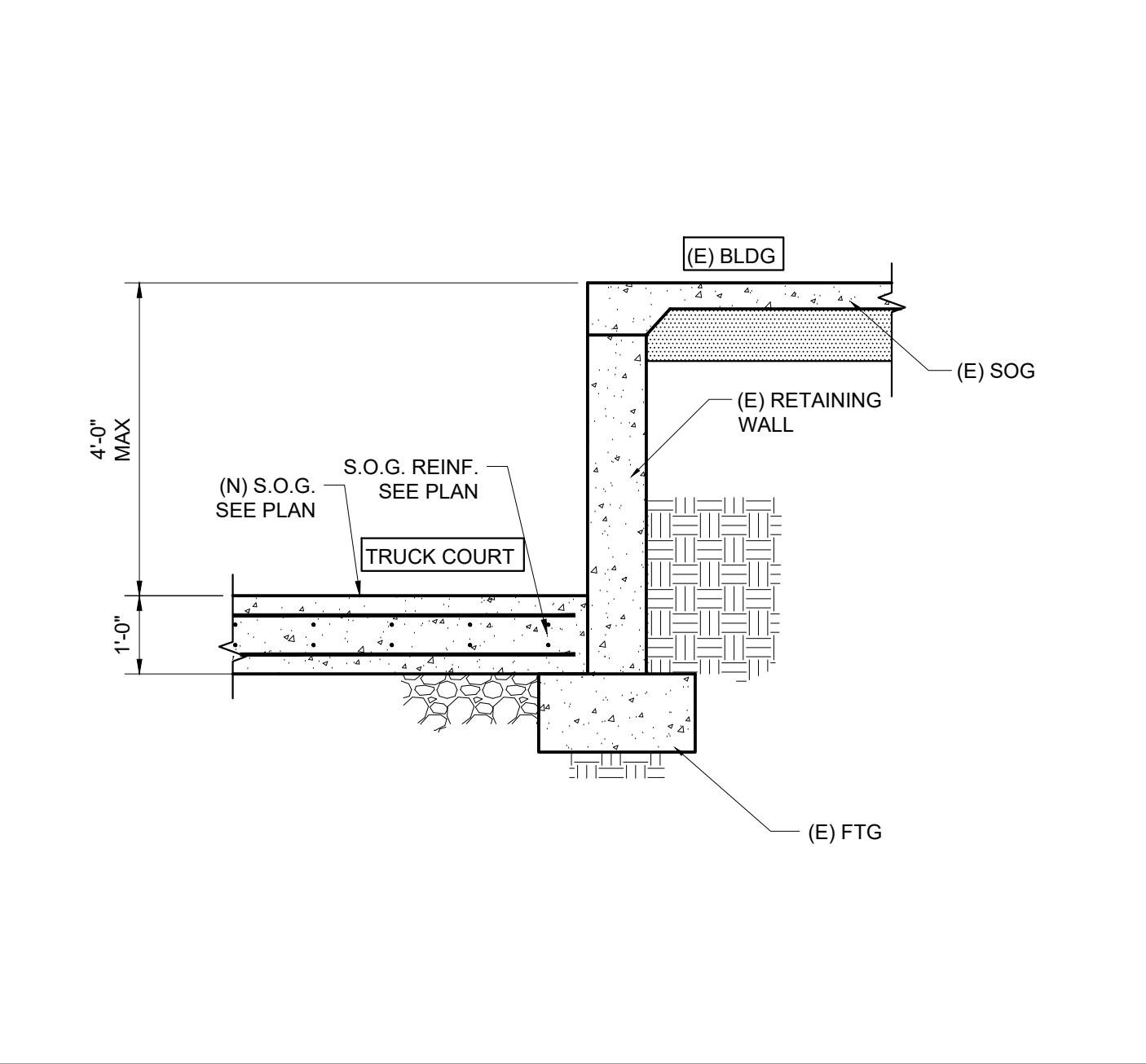
8 DETAIL

scale 3/4"=1'-0"



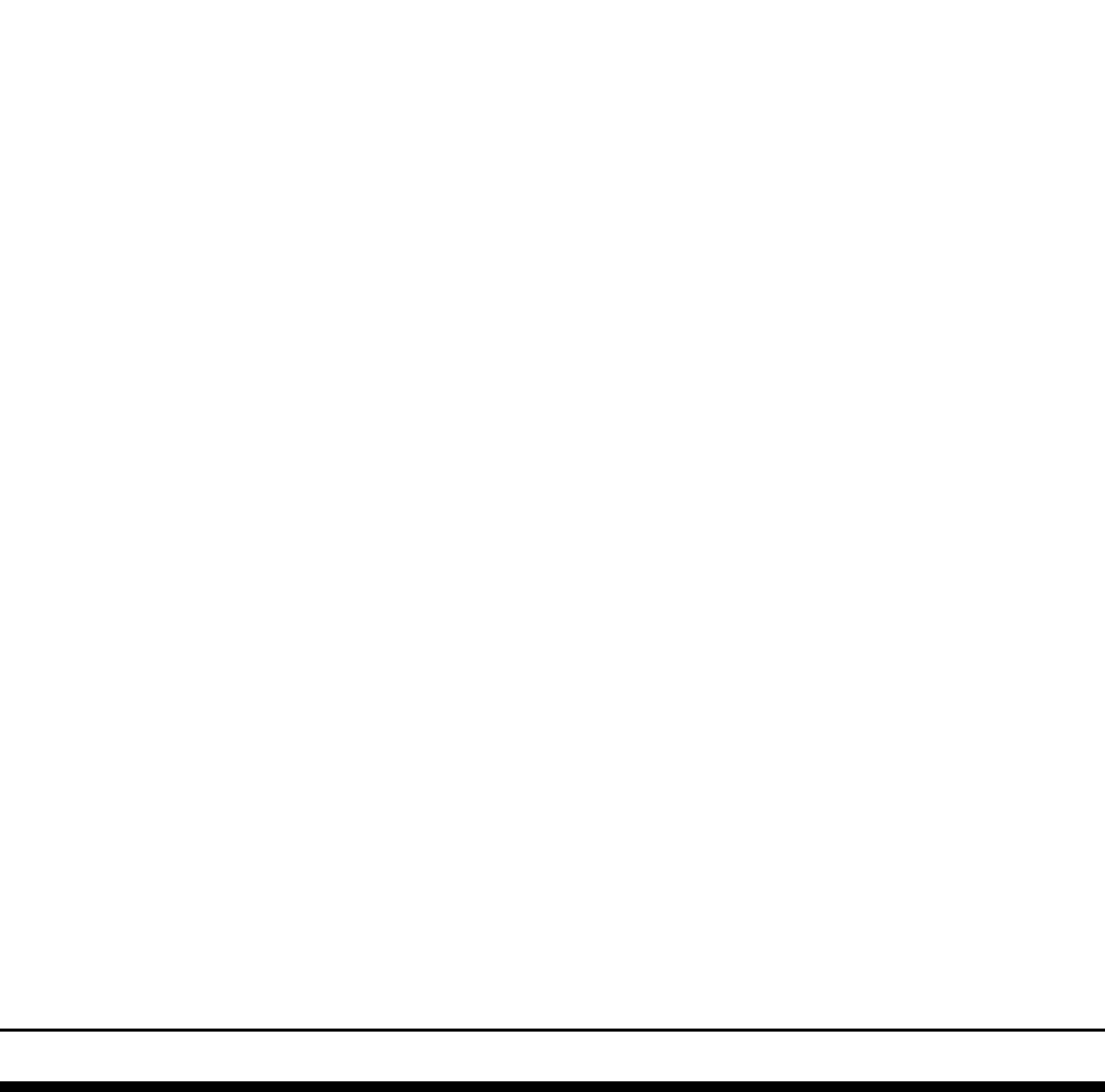
9 DETAIL

scale 3/4"=1'-0"



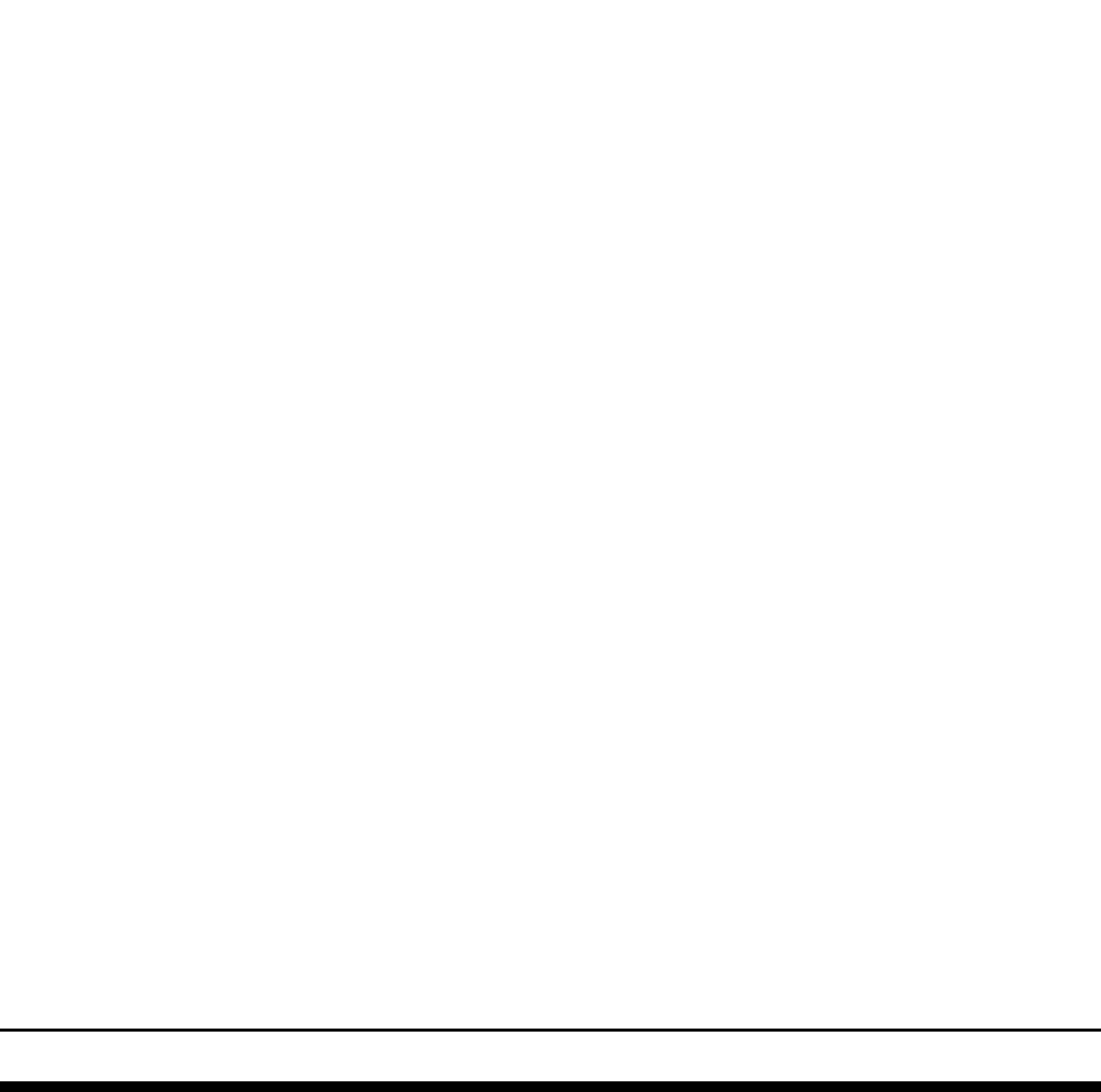
11 DETAIL

scale 3/4"=1'-0"



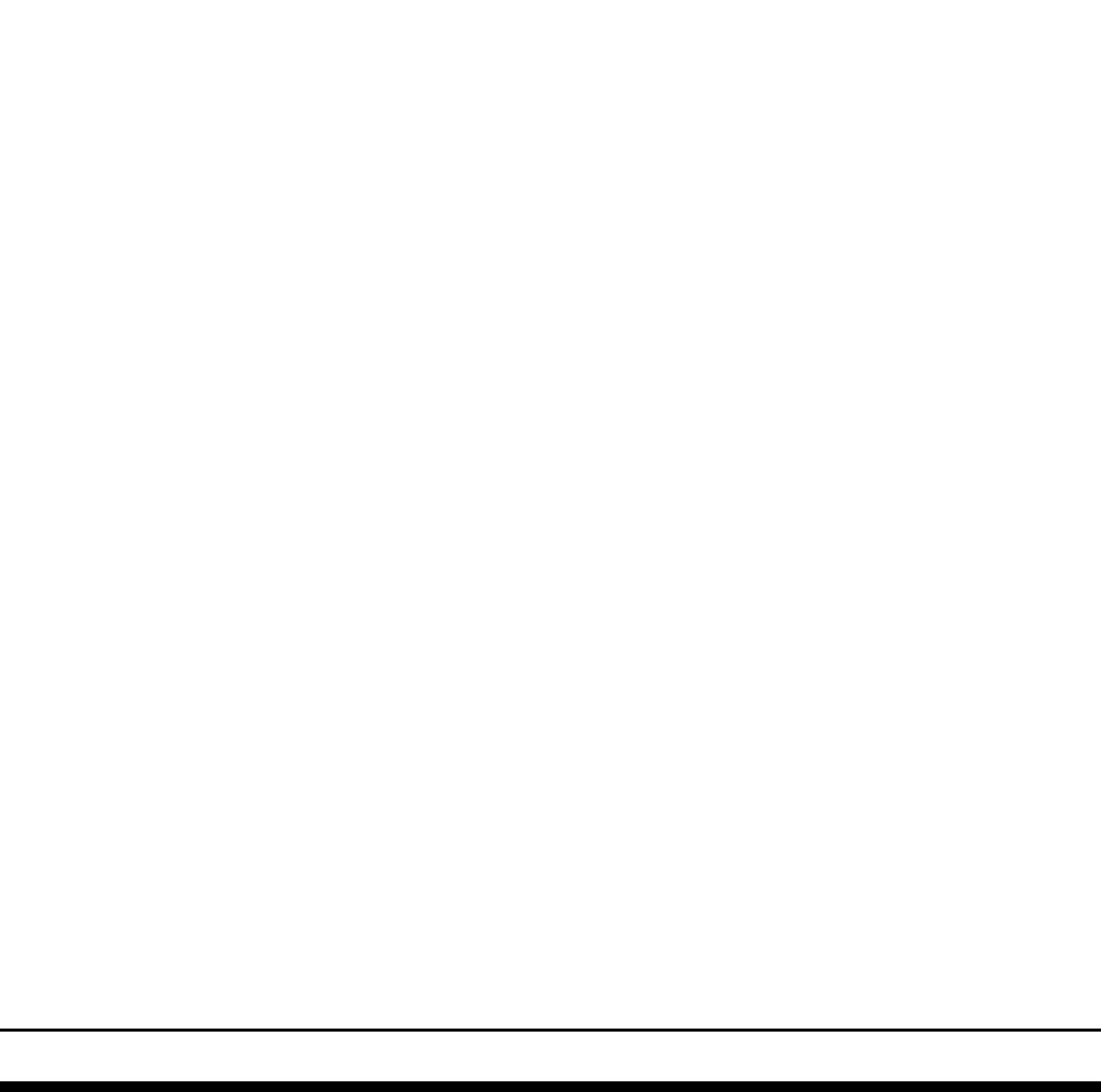
13 DETAIL

scale 3/4"=1'-0"



14 DETAIL

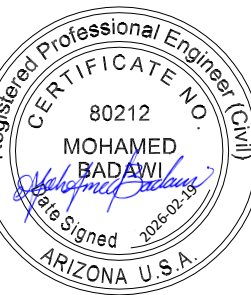
scale 3/4"=1'-0"



1BSE LOGO 4.png

BURKE
STRUCTURAL
ENGINEERS, PC
151 KALMUS DRIVE,
BLDG. E-140
COSTA MESA, CA. 92626
(657) 289-0460

STAMP



REVISION SCHEDULE

DELTA ISSUED BY ISSUE DATE

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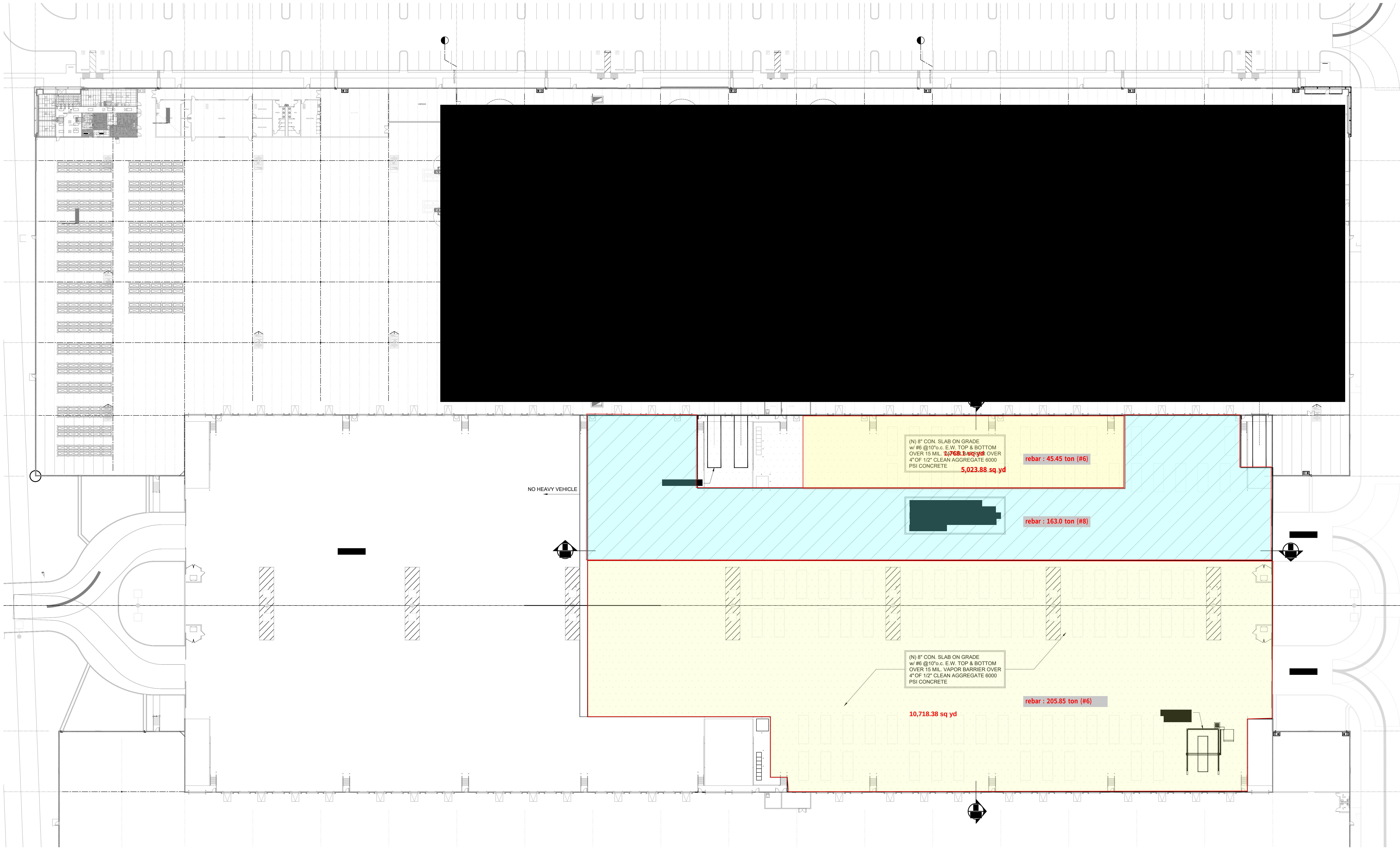
TYPICAL CONCRETE DETAILS

HACE, INC

TENANT IMPROVEMENT WEST 202 LOGISTICS
675-555 N. 55th AVE, PHOENIX, AZ 85043

DATE	2026-02-10
SCALE	AS SHOWN
PROJECT MANAGER	MEB
CHECKED BY	MEB
DRAWN BY	NMN
JOB NO.	26011
SHEET	

S501



(N) 8" CON. SLAB ON GRADE
w/ #6 @10"o.c. E.W. TOP & BOTTOM
OVER 15 MIL. VAPOR BARRIER OVER
4" OF 1/2" CLEAN AGGREGATE 6000
PSI CONCRETE
5,023.88 sq yd

rebar : 45.45 ton (#6)

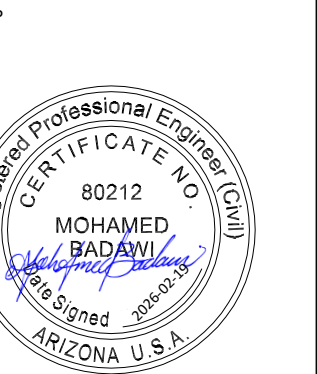
rebar : 163.0 ton (#8)

(N) 8" CON. SLAB ON GRADE
w/ #6 @10"o.c. E.W. TOP & BOTTOM
OVER 15 MIL. VAPOR BARRIER OVER
4" OF 1/2" CLEAN AGGREGATE 6000
PSI CONCRETE

rebar : 205.85 ton (#6)

10,718.38 sq yd

- STRADDLE CARRIER ONLY
- HEAVY TRAFFIC



REVISION SCHEDULE		
DELTA	ISSUED BY	ISSUE DATE

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CONCRETE SLAB PLAN

HACE, INC
TENANT IMPROVEMENT WEST 202 LOGISTICS
675-555 N. 55th AVE, PHOENIX, AZ 85043

DATE	2026-02-10
SCALE	AS SHOWN
PROJECT MANAGER	MEB
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JOB NO.	26011
SHEET	